MYP 5 · MATHEMATICS

Worked Answers
and Mark Scheme

MYP 5 eAssessment-Style Question Bank

Criteria B, C and D · Complete solutions with mark allocation

Investigating Patterns B
generalise, test, justify

Communicating C
represent, reason, explain

Applying Mathematics D
model, calculate, evaluate

43	537	100%
tasks covered	marks allocated	worked solutions

Teacher and student edition · Answers aligned to the printed question bank

Prepared October 2026 · 56-page question bank · Sections 2 to 13

How to use this answer sheet

This document is the companion to the 56-page MYP 5 Mathematics eAssessment-style question bank. It gives a complete worked solution, a boxed final answer and a short marking note for every part of every task — 43 tasks and 537 marks in total.

What each block contains

Part	The part label exactly as it appears in the question bank.
Worked solution	A model method with the reasoning written out — the standard a Criterion C answer should reach.
Answer	The final result, boxed, with units where they matter.
Marking	How the marks in the right-hand column are earned, and where candidates usually lose them.
Examiner note	The insight that separates a secure answer from a routine one — the difference between level 6 and level 8.

Three ways to use it

Student, after attempting. Cover the answer boxes with a sheet of paper, compare methods, then read the marking note to see where the marks sat.

Teacher, when marking. The marking note gives the split for each part; the total in the header matches the question bank exactly.

Revision. Work backwards from the examiner notes: they list the mistakes that most often cost marks in each topic.

Assessment blueprint

Sec.	Focus	Criterion	Tasks	Marks	Share
2	Investigating Patterns	Criterion B	B1, B2, B3, B4, B5, B6	92	17.1%
3	Communicating	Criterion C	C1, C2, C3, C4, C5, C6, C7, C8	97	18.1%
4	Applying Mathematics in Real-Life Contexts	Criterion D	D1, D2, D3, D4, D5, D6, D7, D8, D9, D10	139	25.9%
5	Sequences and Series in Context	Sequences & Patterns	S1, S2	20	3.7%
6	Equations, Inequalities and Fractions	Algebra	A1, A2	24	4.5%
7	Modelling with Quadratics	Quadratic Relationships	Q1	14	2.6%
8	Trigonometric Models	Functions & Graphs	F1	14	2.6%
9	Circle Theorems, Solids and Scale	Geometry & Measurement	G1, G2, G3	35	6.5%
10	Working with Grouped Data	Statistics	St1	12	2.2%
11	Tree Diagrams Without Replacement	Probability	P1	12	2.2%
12	Depreciation and Reducing Balance	Financial Mathematics	Fn1	12	2.2%
13	MYP 5 eAssessment Challenges	Mixed Challenge	M1, M2, M3, M4	66	12.3%
	Total		43 tasks	537	100%

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Page numbers refer to this answer sheet, not to the question bank. The task codes (B1, C4, D7, M2 …) are identical in both documents.

CRITERION B

Investigating Patterns

2

Every answer must move from *observation* to *generalisation* to *justification*. A formula alone earns about half the marks: the rest come from describing the pattern in words, testing it and justifying it.

CRITERION B

14 MARKS

B1 Matchstick Triangles

Difficulty: Applying · Equilateral triangles are built in a row; each new triangle shares one stick with the previous one.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(a) Count the sticks in the four printed figures, then continue the pattern: 3, 5, 7, 9, 11, 13 — add 2 for each new triangle. **3**

Figure, n	1	2	3	4	5	6
Matchsticks, M	3	5	7	9	11	13

Answer **M = 3, 5, 7, 9, 11, 13**

Marking *1 mark for two correct values · 2 marks for one slip · 3 marks for all six.*

(b) Each extra triangle adds **two** sticks, so the first differences are constant (+2). Reason: a new triangle shares one of its sides with the triangle before it, so only two of its three sides are new. **2**

Answer **The number of sticks increases by 2 each time (constant first difference).**

Marking *1 mark for "+2" · 1 mark for linking it to the shared side.*

(c) Start at 3 sticks and add 2 sticks (n − 1) times: **3**

$M(n) = 3 + 2(n - 1) = 3 + 2n - 2$

$M(n) = 2n + 1$

Answer **M(n) = 2n + 1**

Marking *1 mark for 3 + 2(n − 1) · 1 mark for simplifying · 1 mark for correct notation.*

(d) Substitute n = 20 into the general rule: **2**

$M(20) = 2(20) + 1 = 40 + 1 = 41$

Answer **41 matchsticks**

Marking *1 mark for substitution · 1 mark for the answer with units.*

CRITERION B

14 MARKS

B1 Matchstick Triangles

Difficulty: Applying · Equilateral triangles are built in a row; each new triangle shares one stick with the previous one.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(e)

Set M(n) = 100: 2n + 1 = 100 so 2n = 99 so n = 49.5.

A figure number must be a whole number, and 49.5 is not.

Check either side: Figure 49 uses 99 sticks and Figure 50 uses 101 sticks, so 100 is skipped.

Answer **No — 100 is not of the form 2n + 1 for a whole number n (99 and 101 are possible).**

Marking *1 mark for forming and solving 2n + 1 = 100 · 1 mark for a justified conclusion.*

2

(f)

The first triangle needs 3 sticks. Every triangle after the first adds 2 new sticks, because one of its sides is already counted as the right-hand side of the previous triangle.

There are (n − 1) such extra triangles, so M(n) = 3 + 2(n − 1).

This is a structural justification: it explains *why* the rule works for every n, not just for the figures printed.

Answer **3 + 2(n − 1): one shared side per new triangle gives the constant +2.**

Marking *1 mark for identifying the shared side · 1 mark for a general argument covering all n.*

2

EXAMINER NOTE *A table alone is not a justification. Criterion B rewards the move from "I noticed" to "this is why it must always be true".*

CRITERION B CRITERION C

15 MARKS

B2 Square Dot Arrays

Difficulty: Reasoning · Figure k is a square array with k dots along each side.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(a)

Total dots: each side has n dots, so the array is n × n.

Perimeter dots: count the four sides (4n) and correct the four corner dots counted twice (−4).

For n = 1 the single dot is the whole perimeter, so P(1) = 1.

n	1	2	3	4	5	6
Total dots, D(n)	1	4	9	16	25	36
Perimeter dots, P(n)	1	4	8	12	16	20

Answer **D: 1, 4, 9, 16, 25, 36 and P: 1, 4, 8, 12, 16, 20**

Marking *1 mark for the D row · 2 marks for the P row (1 mark if only P(4) onwards is right).*

4

(b)

D(n) = n × n, since Figure n is an n by n square of dots.

P(n) = 4n − 4, because the four sides contribute 4n dots but each of the four corners has been counted twice.

Answer **$D(n) = n^2$ and P(n) = 4n − 4 (n ≥ 2)**

Marking *1 mark for $D(n) = n^2$ · 2 marks for P(n) = 4n − 4 with the "−4 corners" reason.*

3

CRITERION B CRITERION C

15 MARKS

B2 Square Dot Arrays

Difficulty: Reasoning · Figure k is a square array with k dots along each side.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(c)

D(n): 1 4 9 16 25 36
1st diff: 3 5 7 9 11
2nd diff: 2 2 2 2
The second differences are constant, which is the signature of a quadratic sequence.

Answer **Constant second difference of 2, so D(n) is quadratic.**

Marking *1 mark for a correct difference table · 1 mark for the conclusion from constant 2nd differences.*

2

(d)

$I(n) = D(n) - P(n) = n^2 - (4n - 4)$
$I(n) = n^2 - 4n + 4 = (n - 2)^2$

Answer **$I(n) = n^2 - 4n + 4 = (n - 2)^2$**

Marking *1 mark for substituting · 1 mark for simplifying · 1 mark for the factorised form.*

3

(e)

Set $I(n) = P(n)$: $(n - 2)^2 = 4n - 4$
$n^2 - 4n + 4 = 4n - 4$
$n^2 - 8n + 8 = 0$
$n = [8 \pm \sqrt{(64 - 32)}] / 2 = 4 \pm 2\sqrt{2} \rightarrow n \approx 1.17$ or $n \approx 6.83$
Neither value is a whole number, so no numbered figure has equal interior and perimeter dots; the two counts pass through equality between Figures 6 and 7.
Sanity check at n = 7: I = 25, P = 24 — interior overtakes perimeter here.

Answer **No whole-number solution; equality would occur at n ≈ 6.83 (between Figures 6 and 7).**

Marking *1 mark for a correct equation · 1 mark for solving · 1 mark for interpreting the non-integer result.*

3

EXAMINER NOTE *Part (e) is deliberately non-integer: a strong answer interprets the result rather than forcing n = 7. Accept "no solution in whole numbers" with working.*

CRITERION B

16 MARKS

B3 Staircase Block Patterns

Difficulty: Investigating · Staircases are built from unit cubes in columns of heights n, n − 1, ..., 1.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(a)

Read each figure column by column: Figure 1 has 1 cube, Figure 2 has 1 + 2, Figure 3 has 1 + 2 + 3, and so on.

n	1	2	3	4	5	6
Cubes, S(n)	1	3	6	10	15	21

Answer **S: 1, 3, 6, 10, 15, 21**

Marking *1 mark per two correct entries; 3 marks for an accurate table.*

3

CRITERION B

16 MARKS

B3 Staircase Block Patterns

Difficulty: Investigating · Staircases are built from unit cubes in columns of heights n, n − 1, …, 1.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	S: 1 3 6 10 15 21 1st diff: 2 3 4 5 6 2nd diff: 1 1 1 1 The first differences increase by 1 each time and the second differences are constant, so the sequence is quadratic (triangular numbers). **Answer** **Quadratic sequence — constant second difference of 1.** **Marking** *1 mark for first differences · 1 mark for second differences · 1 mark for the classification.*	**3**
(c)	Let $S(n) = an^2 + bn + c$ and substitute three values: n = 1: a + b + c = 1 (i) n = 2: 4a + 2b + c = 3 (ii) (ii) − (i): 3a + b = 2 (iv) n = 3: 9a + 3b + c = 6 (iii) (iii) − (ii): 5a + b = 3 (v) (v) − (iv): 2a = 1 → a = ½, then b = ½, then c = 0. $S(n) = \frac{1}{2}n^2 + \frac{1}{2}n = n(n+1)/2$ **Answer** **$S(n) = \frac{1}{2}n^2 + \frac{1}{2}n$, i.e. $S(n) = n(n+1)/2$** **Marking** *2 marks for forming equations · 1 mark for solving a, b · 1 mark for c and the final rule.*	**4**
(d)	$S(30) = \frac{1}{2}(30)^2 + \frac{1}{2}(30) = 450 + 15 = 465$. **Answer** **465 cubes** **Marking** *1 mark for substitution · 1 mark for the answer.*	**2**
(e)	Test the claim $T(n) = n(n+1)(n+2)/6$: T(1) = 1·2·3/6 = 1 = S(1) ✓ T(2) = 2·3·4/6 = 4, but S(2) = 3 ✗ One counter-example is enough to disprove a general claim. Algebraically T(n) = S(n)(n + 2)/3, which equals S(n) only when (n + 2)/3 = 1, i.e. n = 1. For every n ≥ 2, T(n) is strictly larger. T(n) is in fact the tetrahedral number — the sum of triangular numbers, not the triangular number itself. **Answer** **Disproved: T(2) = 4 ≠ 3; T(n) = S(n)(n + 2)/3, so they agree only at n = 1.** **Marking** *2 marks for checking n = 1 to 4 · 1 mark for the counter-example · 1 mark for the algebraic comparison.*	**4**

EXAMINER NOTE *Part (e) rewards disproving by counter-example — an important Criterion B skill. Students who "prove" the claim usually expand both expressions carelessly.*

CRITERION B | CRITERION C

18 MARKS

B4 Hexagonal Flower

Difficulty: Extension Challenge · A central dot is surrounded by hexagonal rings of dots.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(a) Ring 1 is the centre itself (1 dot). Ring 2 has 6 dots, ring 3 has 12, ring 4 has 18, ring 5 has 24 — each ring adds 6 more than the one before. **4**

n	1	2	3	4	5
Dots added in ring	1	6	12	18	24
Total, H(n)	1	7	19	37	61

Answer **H: 1, 7, 19, 37, 61 (additions 1, 6, 12, 18, 24)**

Marking *2 marks for the "added" row · 2 marks for the totals.*

(b) Each hexagonal ring has six sides, and side $k-1$ of ring k carries $(k-1)$ dots. **3**
So ring k ($k \geq 2$) contains $6(k-1)$ dots.

Answer **Ring k has $6(k-1)$ dots ($k \geq 2$); ring 1 is the single centre dot.**

Marking *1 mark for "six sides" · 2 marks for the formula $6(k-1)$.*

(c) $H(n) = 1 + 6(1) + 6(2) + \ldots + 6(n-1)$ **5**
$H(n) = 1 + 6[1 + 2 + \ldots + (n-1)]$
The bracket is the sum of the first $(n-1)$ integers: $(n-1)n/2$.
$H(n) = 1 + 6 \cdot n(n-1)/2 = 1 + 3n(n-1)$
Check: $n = 4$ gives $1 + 3(4)(3) = 37$ ✓, matching the figure.

Answer **$H(n) = 1 + 3n(n-1)$, derived by summing the six-sided rings.**

Marking *2 marks for the summation set-up · 2 marks for summing the series · 1 mark for the check/justification.*

(d) $H(10) = 1 + 3(10)(9) = 1 + 270 = 271$. **2**

Answer **271 dots**

Marking *2 marks (1 for substitution, 1 for the value).*

(e) $1 + 3n(n-1) = 217 \rightarrow 3n(n-1) = 216 \rightarrow n(n-1) = 72$ **4**
$n^2 - n - 72 = 0 \rightarrow (n-9)(n+8) = 0 \rightarrow n = 9$ ($n = -8$ rejected).

Answer **9 rings**

Marking *2 marks for a correct equation · 1 mark for solving · 1 mark for rejecting the negative root.*

EXAMINER NOTE *The strength of the answer to (c) is the* ***summation argument****: the ring sizes form an arithmetic sequence, so their total is quadratic in n.*

CRITERION B | CRITERION C

15 MARKS

B5 Arithmetic vs Geometric Growth

Difficulty: Reasoning · Sequence A: first term 20, common difference 15. Sequence G: first term 2, common ratio 2.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(a) Arithmetic: add 15 each time. Geometric: multiply by 2 each time. **3**

n	1	2	3	4	5	6
A(n)	20	35	50	65	80	95
G(n)	2	4	8	16	32	64

Answer **A: 20, 35, 50, 65, 80, 95 G: 2, 4, 8, 16, 32, 64**

Marking *2 marks for A · 1 mark for G.*

(b) $A(n) = 20 + 15(n - 1) = 15n + 5$
$G(n) = 2 \times 2^{n-1} = 2^n$ **3**

Answer **$A(n) = 15n + 5$ and $G(n) = 2^n$**

Marking *2 marks for A(n) · 1 mark for $G(n) = 2^n$.*

(c) $2^n > 10\,000$. Powers of 2: $2^{13} = 8192$, $2^{14} = 16\,384$. **2**
So the 14th term is the first to exceed 10 000. ($\log_2 10\,000 \approx 13.29$ confirms this.)

Answer **n = 14**

Marking *1 mark for a correct strategy (listing or logs) · 1 mark for n = 14.*

(d) The arithmetic sequence plots as a **straight line** with gradient 15 (constant difference). **4**
The geometric sequence plots as an **increasing curve** that starts below the line, crosses it, and then rises far more steeply.
Both graphs begin near the origin; G is below A until $n \approx 7$, then above for all later n.

Answer **Straight line (constant difference) vs. upward curve (constant ratio); G overtakes A near n = 7.**

Marking *1 mark for two correct graphs · 2 marks for describing difference in shape · 1 mark for the crossing point.*

(e) Compare 2^n with $15n + 5$: **3**
$n = 6$: $64 < 95$ $n = 7$: $128 > 110$ → the smallest n is 7.
Once G(7) > A(7), the gap cannot close: each step multiplies G by 2 while A only adds 15. The ratio $G/A = 2^n/(15n + 5)$ grows without bound, so G stays ahead for all larger n.

Answer **n = 7; exponential growth eventually dominates any linear growth.**

Marking *1 mark for testing values · 1 mark for n = 7 · 1 mark for the growth-rate justification.*

EXAMINER NOTE *Answers to (e) that only say "the graph is steeper" earn 1 of 3 marks. The key idea is the ratio between the terms, not a single comparison.*

CRITERION B | CRITERION C

14 MARKS

B6 Transforming $y = x^2$

Difficulty: Investigating · Investigate how changing the equation of $y = x^2$ transforms its graph.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	(i) $y = x^2 + 3$: translate **3 units up**; vertex (0, 3). (ii) $y = (x + 2)^2$: translate **2 units left** (inside the bracket acts the opposite way); vertex (−2, 0). (iii) $y = -x^2$: **reflection in the x-axis**; vertex (0, 0), now a maximum. (iv) $y = 2x^2$: **vertical stretch, factor 2** (narrower); vertex (0, 0). **Answer** **up 3 · left 2 · reflected in x-axis · stretched ×2 vertically; vertices (0,3), (−2,0), (0,0), (0,0)** **Marking** *1 mark per transformation and vertex (4 marks).*	**4**
(b)	Evaluate both sides at three values of x: $x = 0$: $(0 - 3)^2 + 1 = 10$ and $0 - 0 + 10 = 10$ ✓ $x = 1$: $(-2)^2 + 1 = 5$ and $1 - 6 + 10 = 5$ ✓ $x = 2$: $(-1)^2 + 1 = 2$ and $4 - 12 + 10 = 2$ ✓ Three checks support the claim, but the rigorous proof is the expansion: $(x - 3)^2 + 1 = x^2 - 6x + 9 + 1 = x^2 - 6x + 10$. **Answer** **The two expressions agree at every x (identity), shown by expanding.** **Marking** *2 marks for three correct substitutions · 1 mark for recognising that expansion is the proof.*	**3**
(c)	$y = x^2 - 4x + 7 = (x^2 - 4x) + 7 = (x - 2)^2 - 4 + 7$ $y = (x - 2)^2 + 3$, so the vertex is (2, 3). **Answer** **$y = (x - 2)^2 + 3$, vertex (2, 3)** **Marking** *1 mark for halving the coefficient · 1 mark for completing the square · 1 mark for the vertex.*	**3**
(d)	$y = 2(x + 1)^2 - 4$ compared with $y = x^2$: • inside the bracket: shift **1 unit left**; • coefficient 2: **vertical stretch by factor 2**; • −4 outside: shift **4 units down**; • no negative sign, so no reflection. Vertex: (−1, −4); it is a minimum, and the curve is narrower than $y = x^2$. **Answer** **Left 1, vertical stretch ×2, down 4; vertex (−1, −4), minimum.** **Marking** *1 mark per correct transformation (3) · 1 mark for vertex and nature.*	**4**

EXAMINER NOTE *Watch the classic error in (d): the +1 inside the bracket moves the graph* ***left****, not right.*

CRITERION C

Communicating

3

Criterion C is about the *quality of the communication*: notation, logical steps, labelled diagrams, correct units and explanations that a reader can follow. A correct answer with no working rarely reaches the top band.

CRITERION C

10 MARKS

C1 Spot the Error: Solving an Equation

Difficulty: Applying · A student solved 2(x − 3) = 3x + 4 as shown; identify and correct the errors.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	**Step 1 is wrong.** The bracket was not fully expanded: 2(x − 3) = 2x − 6, not 2x − 3. The 2 must multiply both terms inside the bracket. **Step 2 inherits the error** and also mixes up the constant: using the correct expansion, 2x − 6 = 3x + 4 gives −6 − 4 = 3x − 2x. **Step 3 gives −10 = x** (not −7 = x). **Step 4 is a sign error:** −10 = x means x = −10, not x = 7. A negative number cannot become positive by rearranging. **Answer** **Errors at Step 1 (incomplete expansion), Step 2 (wrong constant carried forward) and Step 4 (sign change).** **Marking** *1 mark for each error correctly identified with a reason (max 3) · 1 mark for clear communication.*	**4**
(b)	2(x − 3) = 3x + 4 2x − 6 = 3x + 4 2x − 3x = 4 + 6 −x = 10 x = −10 **Answer** **x = −10** **Marking** *1 mark for correct expansion · 1 mark for correct rearrangement · 1 mark for x = −10.*	**3**
(c)	LHS: 2(−10 − 3) = 2(−13) = −26 RHS: 3(−10) + 4 = −30 + 4 = −26 Both sides equal −26, so the solution is verified. **Answer** **LHS = RHS = −26 ✓** **Marking** *1 mark for substituting into both sides · 1 mark for a clear conclusion.*	**3**

EXAMINER NOTE *This task assesses error analysis, a common eAssessment item type: name the error, explain it, then correct it.*

CRITERION C CRITERION D

12 MARKS

C2 Reading a Linear Graph

Difficulty: Applying · A bicycle-hire graph shows cost C (dollars) against hours t; it passes through (0, 10) with gradient 6.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	The fixed deposit is the cost when t = 0, i.e. the **y-intercept** of the graph: the line crosses the vertical axis at 10. **Answer** **$10 (the y-intercept)** **Marking** *1 mark for reading the intercept · 1 mark for linking it to "cost at zero hours".*	**2**
(b)	Choosing two points on the line, e.g. (0, 10) and (5, 40): gradient = (40 − 10)/(5 − 0) = 30/5 = 6 Units: dollars per hour — the hire charge for each additional hour. **Answer** **Gradient = 6 dollars per hour (cost per hour of hire).** **Marking** *2 marks for the calculation with points · 1 mark for the units and interpretation.*	**3**
(c)	C = mt + b with m = 6 and b = 10. **Answer** **C = 6t + 10** **Marking** *1 mark for correct m and b · 1 mark for the equation.*	**2**
(d)	C(5) = 6(5) + 10 = 30 + 10 = 40. **Answer** **$40** **Marking** *1 mark for substitution · 1 mark for the answer with units.*	**2**
(e)	46 = 6t + 10 → 36 = 6t → t = 6. Check: 6(6) + 10 = 46 ✓ **Answer** **6 hours** **Marking** *2 marks for solving the equation · 1 mark for checking/units.*	**3**

EXAMINER NOTE *Always state units for a gradient in a context question — it is a Criterion C hallmark.*

CRITERION C CRITERION B

14 MARKS

C3 Analysing a Parabola

Difficulty: Reasoning · Consider $f(x) = x^2 - 2x - 3$ and its graph.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	y-intercept: put x = 0 → f(0) = −3. x-intercepts: the curve cuts the x-axis at −1 and 3. Vertex: the lowest point is at x = 1, y = −4. **Answer** **(0, −3); (−1, 0) and (3, 0); (1, −4)** **Marking** *1 mark for y-intercept · 2 marks for both x-intercepts · 1 mark for the vertex.*	**4**

CRITERION C | CRITERION B

14 MARKS

C3 Analysing a Parabola

Difficulty: Reasoning · Consider $f(x) = x^2 - 2x - 3$ and its graph.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	$x^2 - 2x - 3$: two numbers with product -3 and sum -2 are -3 and $+1$. $x^2 - 2x - 3 = (x - 3)(x + 1)$ Roots $x = 3$ and $x = -1$, confirming the graph. **Answer** **$f(x) = (x - 3)(x + 1)$, so $x = 3$ or $x = -1$** **Marking** *2 marks for the factorisation · 1 mark for stating the roots.*	**3**
(c)	$x^2 - 2x - 3 = (x^2 - 2x + 1) - 1 - 3$ $= (x - 1)^2 - 4$ Vertex form $y = (x - 1)^2 - 4$ gives vertex $(1, -4)$ — the same as the graph. **Answer** **$f(x) = (x - 1)^2 - 4$, vertex $(1, -4)$** **Marking** *1 mark for $(x - 1)^2$ · 1 mark for adjusting the constant · 1 mark for the vertex.*	**3**
(d)	The coefficient of x^2 is $+1 > 0$, so the parabola opens upwards: the vertex is the lowest point, hence a **minimum**. That minimum value is -4 at $x = 1$. **Answer** **Minimum of -4 (at $x = 1$), because $a = 1 > 0$.** **Marking** *1 mark for "minimum" · 1 mark for the reason based on the sign of a.*	**2**
(e)	Axis of symmetry: $x = 1$. From the factored form: the axis lies midway between the roots, $x = (-1 + 3)/2 = 1$. From the vertex form: the axis is $x = h$ in $y = (x - h)^2 + k$, so $x = 1$. **Answer** **$x = 1$, found both as the midpoint of the roots and as $x = h$ in vertex form.** **Marking** *1 mark for $x = 1$ · 1 mark for both derivations.*	**2**

EXAMINER NOTE *Comparing three representations (graph, factorised, vertex form) is exactly the kind of Criterion C "move between forms" the eAssessment rewards.*

CRITERION C | CRITERION B

13 MARKS

C4 Exponential Growth vs Decay

Difficulty: Reasoning · The graphs of $y = 2$ and $y = (½)$ are shown.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	$y = 2^x$ has base $2 > 1$, so each step multiplies by 2 → **growth**, curve rising to the right. $y = (½)^x$ has base $0 < ½ < 1$, so each step multiplies by ½ → **decay**, curve falling to the right. Note $(½)^x = 2^{-x}$, a reflection of $y = 2^x$ in the y-axis. **Answer** **2^x is growth (base > 1); $(½)^x$ is decay (0 < base < 1).** **Marking** *1 mark for each identification · 1 mark for the reasoning from the base.*	**3**

CRITERION C | CRITERION B

13 MARKS

C4 Exponential Growth vs Decay

Difficulty: Reasoning · The graphs of $y = 2$ and $y = (½)$ are shown.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	At $x = 0$, $a^0 = 1$ for every $a \neq 0$, so both graphs pass through $(0, 1)$. They cannot pass through $(0, 0)$, because no positive base raised to 0 gives 0. **Answer** **$(0, 1)$ for both; every $y = a^x$ has y-intercept 1 because $a^0 = 1$.** **Marking** *1 mark for the intercept · 2 marks for the explanation.*	**3**
(c)	$2^x = 16 = 2^4 \rightarrow x = 4$. $(½)^x = 8 \rightarrow 2^{-x} = 2^3 \rightarrow -x = 3 \rightarrow x = -3$. **Answer** **$x = 4$ and $x = -3$** **Marking** *2 marks for $2^x = 16$ · 1 mark for $(½)^x = 8$.*	**3**
(d)	Both graphs approach the x-axis but never touch it: the horizontal asymptote is $y = 0$. In context, an exponentially decaying quantity never quite reaches zero — it keeps halving. **Answer** **$y = 0$; the value becomes arbitrarily small but never zero.** **Marking** *1 mark for the asymptote · 1 mark for the meaning.*	**2**
(e)	For every real x, $2^x > 0$. A positive number raised to any power is positive, and the graph never reaches the axis, so $2^x = 0$ has no real solution. **Answer** **No real solution, since $2^x > 0$ for all x.** **Marking** *1 mark for "always positive" · 1 mark for a complete explanation.*	**2**

EXAMINER NOTE *The link $(½)^x = 2^{-x}$ turns a "two separate graphs" question into one idea — worth stating explicitly.*

CRITERION C

14 MARKS

C5 Understanding a Rational Function

Difficulty: Investigating · Consider $f(x) = (x - 1)/(x + 2)$.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	The function is undefined where the denominator is zero: $x + 2 = 0 \rightarrow x = -2$. As x approaches -2 the numerator approaches -3 (non-zero) while the denominator approaches 0, so $\lvert f(x) \rvert$ grows without bound. Vertical asymptote: $x = -2$. **Answer** **$x = -2$, because the denominator vanishes there and the numerator does not.** **Marking** *1 mark for $x = -2$ · 2 marks for the explanation.*	**3**

CRITERION C

14 MARKS

C5 Understanding a Rational Function

Difficulty: Investigating · Consider f(x) = (x − 1)/(x + 2).

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	Numerator and denominator have the same degree (both linear), so for large |x| the function behaves like the ratio of leading coefficients: x/x = 1. Horizontal asymptote: y = 1. **Answer** **y = 1 (ratio of the leading coefficients).** **Marking** *1 mark for y = 1 · 2 marks for the reasoning.*	**3**
(c)	x-intercept: numerator zero → x − 1 = 0 → x = 1, giving (1, 0). y-intercept: x = 0 → f(0) = −1/2, giving (0, −½). **Answer** **(1, 0) and (0, −½)** **Marking** *2 marks for the x-intercept · 1 mark for the y-intercept.*	**3**
(d)	(x − 1)/(x + 2) = 3, x ≠ −2 x − 1 = 3(x + 2) = 3x + 6 −7 = 2x x = −3.5 Check: (−3.5 − 1)/(−3.5 + 2) = (−4.5)/(−1.5) = 3 ✓ **Answer** **x = −3.5** **Marking** *1 mark for clearing the fraction · 1 mark for solving · 1 mark for the check.*	**3**
(e)	Critical values: x = −2 (asymptote) and x = 1 (zero). Test each interval: x < −2, e.g. x = −3: (−4)/(−1) = 4 > 0 → positive −2 < x < 1, e.g. x = 0: (−1)/(2) = −½ < 0 → negative x > 1, e.g. x = 2: (1)/(4) > 0 → positive **Answer** **f(x) > 0 for x < −2 or x > 1.** **Marking** *1 mark for the critical values · 1 mark for a correct sign table/answer.*	**2**

EXAMINER NOTE *A sign table is the clearest way to communicate part (e) — exactly what Criterion C is looking for.*

CRITERION C

12 MARKS

C6 The Unit Circle and Sine/Cosine

Difficulty: Reasoning · A point on the unit circle makes an angle θ with the positive x-axis.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	The diagram forms a right-angled triangle with the radius as hypotenuse. The radius of the unit circle is 1, the horizontal leg is cos θ and the vertical leg is sin θ. By Pythagoras: $(\cos\theta)^2 + (\sin\theta)^2 = 1^2$, i.e. $\cos^2\theta + \sin^2\theta = 1$. **Answer** **Pythagoras on the right triangle with hypotenuse 1.** **Marking** *1 mark for identifying the triangle · 2 marks for the complete argument.*	**3**

CRITERION C

12 MARKS

C6 The Unit Circle and Sine/Cosine

Difficulty: Reasoning · A point on the unit circle makes an angle θ with the positive x-axis.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	Take an equilateral triangle of side 2 and drop an altitude: it bisects the base and the 60° angle, giving a right triangle with hypotenuse 2, short leg 1 and altitude $\sqrt{(4-1)} = \sqrt{3}$. Half of the equilateral triangle gives the 30° angle: $\sin 30° = 1/2$, $\cos 30° = \sqrt{3}/2$ The other angle is 60°: $\sin 60° = \sqrt{3}/2$, $\cos 60° = 1/2$ **Answer** **$\sin 30° = ½$, $\cos 30° = \sqrt{3}/2$, $\sin 60° = \sqrt{3}/2$, $\cos 60° = ½$** **Marking** *2 marks for the triangle justification · 2 marks for the four exact values.*	**4**
(c)	Reflecting the point $(\cos θ, \sin θ)$ in the y-axis gives $(-\cos θ, \sin θ)$. The reflected point corresponds to the angle $180° - θ$, so $\cos(180° - θ) = -\cos θ$ and $\sin(180° - θ) = \sin θ$. The y-coordinate (height) is unchanged; the x-coordinate changes sign. **Answer** **Reflection in the y-axis keeps the sine and reverses the cosine.** **Marking** *2 marks for the reflection argument · 1 mark for the conclusion.*	**3**
(d)	$150° = 180° - 30°$ lies in the second quadrant, where sine is positive. $\sin 150° = \sin 30° = 1/2$, a positive value. **Answer** **Positive; $\sin 150° = ½$** **Marking** *1 mark for the sign · 1 mark for the exact value.*	**2**

EXAMINER NOTE *"With justification" in (b) means the exact values must come from a triangle or the circle — not from a calculator.*

CRITERION C CRITERION B

12 MARKS

C7 Proving Thales' Theorem

Difficulty: Extension Challenge · A, B, C lie on a circle with centre O; AB is a diameter. Prove that angle ACB = 90°.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	OA, OB and OC are all radii of the same circle, so OA = OB = OC. That is the definition of a circle: every point on it is the same distance from the centre. **Answer** **All three are radii, hence equal.** **Marking** *2 marks for a complete reason (radii of one circle).*	**2**

CRITERION C CRITERION B

12 MARKS

C7 Proving Thales' Theorem

Difficulty: Extension Challenge · A, B, C lie on a circle with centre O; AB is a diameter. Prove that angle ACB = 90°.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	Triangle OAC has OA = OC, so the angles opposite those sides are equal: ∠OAC = ∠OCA. Triangle OBC has OB = OC, so ∠OBC = ∠OCB. A triangle with two equal sides (and therefore two equal base angles) is isosceles. **Answer** **Both triangles have two radii as sides, so both are isosceles.** **Marking** *1 mark per triangle · 1 mark for naming the base angles.*	**2**
(c)	Let ∠OCA = α and ∠OCB = β. In △OAC, the angle sum gives ∠AOC = 180° − 2α. In △OBC, likewise ∠BOC = 180° − 2β. A, O, B are collinear, so ∠AOC + ∠BOC = 180°: (180° − 2α) + (180° − 2β) = 180° 360° − 2(α + β) = 180° → 2(α + β) = 180° → α + β = 90° But ∠ACB = ∠OCA + ∠OCB = α + β, so ∠ACB = 90° **Answer** **α + β = 90°, and ∠ACB = α + β, hence 90°.** **Marking** *1 mark for ∠AOC · 1 mark for ∠BOC · 1 mark for the straight line · 2 marks for the algebra and conclusion.*	**5**
(d)	Converse: if a triangle ABC has ∠ACB = 90°, then AB is a diameter of its circumcircle — equivalently, C lies on the circle with diameter AB. This is the form used to construct right angles and to test whether an angle is right. **Answer** **If ∠ACB = 90°, then A, B, C lie on a circle with AB as diameter.** **Marking** *2 marks for the statement · 1 mark for correct logic (converse swaps hypothesis and conclusion).*	**3**

EXAMINER NOTE *In a proof, every line needs a reason: angle sum, radii, straight line. Marks are awarded for those reasons, not just the final 90°.*

CRITERION C

10 MARKS

C8 Comparing Two Methods

Difficulty: Reasoning · Two students simplify $(x^2 - 9)/(x^2 + 5x + 6)$.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Student A "cancels" x^2 with x^2 and writes $-9/(5x + 6)$. This is invalid: cancelling is only allowed for **common factors** of the whole numerator and denominator. x^2 and -9 are **terms** joined by subtraction, not factors. Cancelling them changes the value of the expression. Numerical check: at x = 4 the original is 7/42 = 1/6, while Student A's expression gives −9/26, which is different. **Answer** **Terms cannot be cancelled across a subtraction — only factors of the whole expression can.** **Marking** *2 marks for the explanation · 1 mark for a numerical or algebraic counter-example.*	**3**

CRITERION C

10 MARKS

C8 Comparing Two Methods

Difficulty: Reasoning · Two students simplify $(x^2 - 9)/(x^2 + 5x + 6)$.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	Numerator: $x^2 - 9 = (x - 3)(x + 3)$ [difference of two squares] Denominator: $x^2 + 5x + 6 = (x + 2)(x + 3)$ [numbers with product 6 and sum 5] $(x - 3)(x + 3) / [(x + 2)(x + 3)] = (x - 3)/(x + 2)$, $x \neq -3$ The common factor $(x + 3)$ cancels because it is a factor of the whole numerator and denominator. **Answer** $(x - 3)/(x + 2)$, with $x \neq -3$ and $x \neq -2$ **Marking** *1 mark per factorisation · 1 mark for cancelling · 1 mark for the restriction.*	**4**
(c)	The original expression is undefined when its denominator is zero: $x^2 + 5x + 6 = 0 \rightarrow (x + 2)(x + 3) = 0 \rightarrow x = -2$ or $x = -3$. Both values must be stated, even though $(x + 3)$ cancels — the original expression still has a hole at $x = -3$. **Answer** $x = -2$ and $x = -3$ **Marking** *1 mark for setting the denominator to zero · 2 marks for both values with the restriction stated.*	**3**

EXAMINER NOTE *The cancelled factor still creates a hole in the graph. Strong answers mention $x \neq -3$ explicitly and distinguish it from the vertical asymptote at $x = -2$.*

CRITERION D

Applying Mathematics in Real-Life Contexts

4

Criterion D answers need four moves: **identify** the relevant information, **select** a model or tool, **apply** it accurately, and **evaluate** the result in context (units, reasonableness, assumptions, limitations).

CRITERION D CRITERION C

13 MARKS

D1 Ladder Safety

Difficulty: Applying · A 5 m ladder leans against a vertical wall at 65° to the horizontal; the recommended angle is 75°.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	The wall height is the side **opposite** the 65° angle, and the ladder is the hypotenuse: $\sin 65° = h/5 \rightarrow h = 5 \sin 65°$ $h = 5 \times 0.9063 = 4.5315...$ **Answer** $h \approx 4.53$ m	**3**

CRITERION D CRITERION C

13 MARKS

D1 Ladder Safety

Difficulty: Applying · A 5 m ladder leans against a vertical wall at 65° to the horizontal; the recommended angle is 75°.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
	Marking *1 mark for choosing sine · 1 mark for substitution · 1 mark for the answer to 2 d.p.*	
(b)	The distance from the wall is the **adjacent** side: $\cos 65° = d/5 \rightarrow d = 5 \cos 65° = 5 \times 0.4226 = 2.1131…$ **Answer** **$d \approx 2.11$ m** **Marking** *1 mark for cosine · 1 mark for the value.*	**2**
(c)	At 75°: $h = 5 \sin 75° = 5 \times 0.9659 = 4.8296…$ Increase: $4.8296 - 4.5315 = 0.2981 \approx 0.30$ m. **Answer** **≈ 4.83 m — about 0.30 m higher than at 65°.** **Marking** *1 mark for the new height · 1 mark for the comparison · 1 mark for the difference.*	**3**
(d)	At 75° the ladder is the hypotenuse and 4.5 m is opposite the angle: $L = 4.5/\sin 75° = 4.5/0.9659 = 4.6587…$ Rounding **up** to the nearest 0.5 m gives 5.0 m. **Answer** **L ≈ 4.66 m, so a 5.0 m ladder is needed.** **Marking** *1 mark for the trig ratio · 1 mark for 4.66 m · 1 mark for rounding up to 5.0 m.*	**3**
(e)	**Assumption:** the wall is vertical, the ground is horizontal and the ladder is rigid and straight, so the situation is a right triangle. **Why longer:** the foot must be kept clear of a flower bed or step, the top must extend above the window sill for a safe handhold, and the 75° angle must be measured on uneven ground — all reduce the usable height, so a longer ladder is needed. **Answer** **Vertical wall / horizontal ground / rigid ladder; allowances for footing, handhold and uneven ground.** **Marking** *1 mark for an assumption · 1 mark for a practical reason.*	**2**

EXAMINER NOTE *State the trig ratio before substituting. Criterion D rewards showing the model explicitly, not only the number.*

CRITERION D CRITERION B

16 MARKS

D2 Population Growth of a City

Difficulty: Reasoning · A city had 250 000 people at the start of 2015 and grows at 3.2% per year, $P(t) = P_0(1 + r)$.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	P_0 is the initial population; r is the annual growth rate as a decimal. **Answer** **$P_0 = 250\,000$ and $r = 0.032$** **Marking** *1 mark each.*	**2**

CRITERION D | CRITERION B

16 MARKS

D2 Population Growth of a City

Difficulty: Reasoning · A city had 250 000 people at the start of 2015 and grows at 3.2% per year, $P(t) = P_0(1 + r)$.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	The start of 2025 is t = 10 years after 2015. $P(10) = 250\,000(1.032)^{10} = 250\,000 \times 1.37024 = 342\,560...$ **Answer** **≈ 343 000 people (342 560 to the nearest person)** **Marking** *1 mark for t = 10 · 1 mark for substitution · 1 mark for the rounded answer.*	**3**
(c)	$250\,000(1.032)^t = 400\,000$ $(1.032)^t = 1.6$ $t = \log 1.6 / \log 1.032 = 0.47000 / 0.03150 = 14.92$ So t = 15 years, i.e. during 2030. Check: 1.032^{14} gives 1.554 → 388 500 (below); 1.032^{15} gives 1.604 → 401 000 (above). **Answer** **2030 (t ≈ 14.9 years)** **Marking** *1 mark for the equation · 2 marks for solving with logs · 1 mark for interpreting the year.*	**4**
(d)	Doubling requires $(1.032)^t = 2$: $t = \log 2 / \log 1.032 = 0.30103 / 0.03150 = 22.006...$ So the population doubles after about 22 years, in 2037. **Answer** **≈ 22 years (2037)** **Marking** *1 mark for the doubling equation · 2 marks for the logs · 1 mark for the rounded year.*	**4**
(e)	**Limitation:** unlimited exponential growth assumes a constant rate for ever; in reality births fall, migration changes, and housing, water and jobs limit growth. **Better model:** logistic (S-shaped) growth, which starts exponentially but levels off at a carrying capacity, or a piecewise model with a decreasing rate. **Answer** **Constant-rate assumption is unrealistic; a logistic model with carrying capacity is better.** **Marking** *1 mark for a limitation · 1 mark for the model · 1 mark for justification.*	**3**

EXAMINER NOTE *"Which year" needs an interpretation, not just t = 14.92. Rounding up matters: the threshold is crossed during year 15.*

CRITERION D | CRITERION C

14 MARKS

D3 Is the Game Fair?

Difficulty: Reasoning · A spinner: Red 90° pays $5, Blue 120° pays $2, Green 60° pays $10, Yellow 90° pays $0. Entry fee $3.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Each probability is the sector angle divided by 360°, since the spinner is equally likely to stop anywhere.	**3**

CRITERION D CRITERION C

14 MARKS

D3 Is the Game Fair?

Difficulty: Reasoning · A spinner: Red 90° pays $5, Blue 120° pays $2, Green 60° pays $10, Yellow 90° pays $0. Entry fee $3.

PART | WORKED SOLUTION AND MARKING GUIDE | MARKS

Colour	Angle	P(colour)	Payout ($)	P × payout
Red	90°	1/4	5	1.25
Blue	120°	1/3	2	0.6667
Green	60°	1/6	10	1.6667
Yellow	90°	1/4	0	0
Total	360°	1	—	3.5833

Answer **1/4, 1/3, 1/6, 1/4 (total 1)**

Marking *1 mark per two correct probabilities · 1 mark for the total of 1.*

(b) **3**

E(payout) = Σ p × payout
= (1/4)(5) + (1/3)(2) + (1/6)(10) + (1/4)(0)
= 1.25 + 0.6667 + 1.6667 + 0
= 3.5833…

Answer **≈ $3.58 per spin**

Marking *2 marks for the products summed · 1 mark for the value.*

(c) **3**

Expected profit to the player = payout − fee = 3.5833 − 3 = +0.5833.
Since the expected profit is positive, the game favours the **player**: on average a player gains about 58 cents per spin.

Answer **≈ +$0.58 per spin — not fair; the player has the advantage.**

Marking *1 mark for the difference · 1 mark for the conclusion · 1 mark for interpretation.*

(d) **3**

For a fair game the fee would equal the expected payout: $3.58.
For the school to make $1 expected profit per game:
fee = 3.5833 + 1 = 4.5833.

Answer **≈ $4.58 (fair price $3.58; school-profit price $4.58)**

Marking *2 marks for the correct price · 1 mark for distinguishing "fair" from "$1 profit".*

(e) **2**

Expected profit per player = 1 (by design), so for 200 players:
200 × 1 = $200.
(Actual profit varies; $200 is the long-run average.)

Answer **≈ $200 expected**

Marking *1 mark for the multiplication · 1 mark for noting it is an expected value.*

EXAMINER NOTE *Read (d) carefully: "fair" means expected profit zero ($3.58), but the school wants $1 profit, so the price must be higher ($4.58).*

CRITERION D CRITERION C

12 MARKS

D4 Left-Handed or Right-Handed?

Difficulty: Applying · 120 students classified by gender and dominant hand: left-handed 8 male, 6 female; right-handed 52 male, 54 female.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Total left-handed = 8 + 6 = 14 out of 120. P(left) = 14/120 = 7/60 ≈ 0.1167 **Answer** **7/60 ≈ 0.117 (11.7%)** **Marking** *1 mark for 14/120 · 1 mark for the decimal/percentage.*	**2**
(b)	Restrict to females (60 students); 6 are left-handed. P(left | female) = 6/60 = 0.1 **Answer** **0.10 (10%)** **Marking** *1 mark for the restricted denominator · 1 mark for the value.*	**2**
(c)	Restrict to left-handed students (14); 8 are male. P(male | left) = 8/14 = 4/7 ≈ 0.571 **Answer** **4/7 ≈ 0.571 (57.1%)** **Marking** *1 mark for 8/14 · 1 mark for simplifying.*	**2**
(d)	Independent means P(left | male) = P(left | female) = P(left). P(left | male) = 8/60 = 0.1333, P(left | female) = 6/60 = 0.1000, P(left) = 0.1167. Alternatively P(male and left) = 8/120 = 0.0667, while P(male) × P(left) = 0.5 × 0.1167 = 0.0583. The values differ, so the events are **not** independent — but the gap is small. **Answer** **Not independent (0.133 vs 0.100 vs 0.117), though only slightly dependent.** **Marking** *1 mark for a valid test · 1 mark for the values · 1 mark for the conclusion.*	**3**
(e)	The direction of the claim is right: 13.3% of males are left-handed compared with 10.0% of females. But the difference is only 3.3 percentage points and rests on just 8 left-handed males out of 60. A difference of that size is well within sampling variation, so the data **suggest** a tendency but do not establish it. Larger samples (and a significance test) would be needed. **Answer** **Directionally supported (13.3% vs 10.0%) but the evidence is weak — too few left-handers to conclude.** **Marking** *1 mark for the percentages · 1 mark for the comparison · 1 mark for evaluating reliability.*	**3**

EXAMINER NOTE *This is a conditional-probability trap: the denominator changes in (b) and (c). Reading the table row/column totals carefully is the skill being assessed.*

CRITERION D | CRITERION C

16 MARKS

D5 Designing a Water Tank

Difficulty: Reasoning · An open-top cylinder must hold 5 m^3. Curved surface costs 4000 CFA/m^2, base 6000 CFA/m^2.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	$V = \pi r^2 h = 5$ with $r = 1$: $h = 5/(\pi \times 1^2) = 1.5915...$ **Answer** $h \approx 1.59$ m **Marking** *1 mark for the formula · 1 mark for substitution · 1 mark for the answer.*	**3**
(b)	With $h = 2$: $5 = \pi r^2(2) \rightarrow r^2 = 5/(2\pi) = 0.7958$ $r = 0.8921...$ **Answer** $r \approx 0.89$ m **Marking** *1 mark for rearranging · 2 marks for the value.*	**3**
(c)	Curved area $= 2\pi rh = 2\pi(0.8921)(2) = 11.210$ $m^2 \rightarrow 11.210 \times 4000 = 44\,840$ CFA Base area $= \pi r^2 = 2.500$ $m^2 \rightarrow 2.500 \times 6000 = 15\,000$ CFA Total ≈ 59 840 CFA **Answer** ≈ 59 840 CFA **Marking** *1 mark for each area · 1 mark for each cost · 1 mark for the total.*	**4**
(d)	Write the cost in terms of r only, using $h = 5/(\pi r^2)$: $C(r) = 4000(2\pi r \cdot 5/(\pi r^2)) + 6000(\pi r^2) = 40\,000/r + 6000\pi r^2$ Testing r = 0.8 to 1.5 m in steps of 0.1 m:	**4**

r (m)	$h = 5/(\pi r^2)$ (m)	Cost (CFA)	Comment
0.8	2.487	62 064	
0.9	1.965	59 713	
1.0	1.592	58 850	minimum
1.1	1.315	59 172	
1.2	1.105	60 477	
1.3	0.942	62 625	
1.4	0.812	65 517	
1.5	0.707	69 078	

Answer $r \approx 1.0$ m (analytic optimum $r \approx 1.02$ m, $h \approx 1.53$ m), cost ≈ 58 800 CFA

Marking *1 mark for the cost function in one variable · 1 mark for a complete table · 1 mark for identifying the minimum · 1 mark for r and h.*

CRITERION D CRITERION C

16 MARKS

D5 Designing a Water Tank

Difficulty: Reasoning · An open-top cylinder must hold 5 m^3. Curved surface costs 4000 CFA/m^2, base 6000 CFA/m^2.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(e)	• **Strength and stability:** a very tall narrow tank needs thicker walls and a strong base; a very wide tank needs more ground. • **Access and maintenance:** a ladder, a lid to keep out dust and mosquitoes, and room to clean it. • **Local conditions:** available ground, delivery of materials, and water pressure for filling. **Answer** **Stability/strength, access/maintenance, footprint and local site conditions.** **Marking** *1 mark per valid practical consideration (max 2).*	**2**

EXAMINER NOTE *The open top matters: there is no top disc in the cost. Students who include $2\pi r^2$ overestimate the cost.*

CRITERION D CRITERION C

12 MARKS

D6 Comparing Test Scores

Difficulty: Reasoning · Box plots for Class A and Class B on the same mathematics test.

(a) Read the five values from each box plot (values are approximate, ±1). **4**

	Minimum	Q1	Median	Q3	Maximum
Class A	42	57	68	80	95
Class B	35	60	73	78	92

Answer **A: 42, 57, 68, 80, 95 B: 35, 60, 73, 78, 92**

Marking *1 mark per correct row (2) · 2 marks for accuracy on all ten values.*

(b) **4**

Centre: Class B has the higher median (73 vs 68), so a typical B student scored 5 marks more.

Spread: the interquartile ranges are A: 80 − 57 = 23 and B: 78 − 60 = 18, so the middle 50% of B is tighter; the ranges are A: 53 and B: 57, so B has the wider overall spread because of its long lower whisker.

Skewness: A is roughly symmetric (median 68 sits about halfway between the quartiles, and the whiskers are similar at 15 and 15). B is negatively skewed: the median is close to Q3 (73 vs 78) and the lower whisker (35 to 60 = 25) is much longer than the upper (78 to 92 = 14).

Answer **B has the higher centre but the smaller IQR; A is symmetric, B is left-skewed with a low tail.**

Marking *2 marks for centre with values · 1 mark for spread · 1 mark for skewness.*

CRITERION D CRITERION C

12 MARKS

D6 Comparing Test Scores

Difficulty: Reasoning · Box plots for Class A and Class B on the same mathematics test.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(c)	Above Q3 the top 25% of students lie. Interpolating within the upper half of each box: Class A: 25% + (80 − 75)/(80 − 68) × 25% ≈ 25% + 10% = 35% Class B: 25% + (78 − 75)/(78 − 73) × 25% ≈ 25% + 15% = 40% **Answer** **Class B — about 40% compared with about 35% in Class A.** **Marking** *1 mark for using the quartiles · 1 mark for a justified choice (either class accepted with reasoning).*	**2**
(d)	The claim is **partly** true. B's median is higher (73 vs 68) and its upper quartile is close, so typical B performance is better. However A reaches higher at the top (Q3 = 80 vs 78; maximum 95 vs 92) and has a higher floor (minimum 42 vs 35). B's lower whisker shows a group of weak scores. A balanced judgement: B is better *on average*, but A is stronger at the top end and has no very low scores — overall performance is comparable. **Answer** **Partly supported: B is better in the middle, but A is stronger at the top and has a higher minimum.** **Marking** *1 mark for a supported judgement · 1 mark for using values from both classes.*	**2**

EXAMINER NOTE *Compare with numbers. "Class B is better because its box is higher" earns little; quoting median, IQR and whisker lengths earns the marks.*

CRITERION D CRITERION B

15 MARKS

D7 Choosing a Savings Account

Difficulty: Reasoning · Amina invests 200 000 CFA for 5 years. Bank X: 4.5% simple. Bank Y: 4% compound annually. Bank Z: 3.8% compounded quarterly.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Simple interest: $A = P(1 + rt)$ $A = 200\,000(1 + 0.045 \times 5) = 200\,000 \times 1.225$ **Answer** **245 000 CFA (interest 45 000 CFA)** **Marking** *1 mark for the formula · 1 mark for substitution · 1 mark for the answer.*	**3**
(b)	Compound interest: $A = P(1 + r)^t$ $A = 200\,000(1.04)^5 = 200\,000 \times 1.216653$ **Answer** **≈ 243 331 CFA** **Marking** *1 mark for the formula · 1 mark for 1.04^5 · 1 mark for the answer.*	**3**
(c)	Bank X pays 245 000 and Bank Y pays 243 331: 245 000 − 243 331 = 1 669 CFA. **Answer** **Bank X pays more, by about 1 670 CFA.** **Marking** *1 mark for the comparison · 1 mark for the difference.*	**2**

CRITERION D CRITERION B

15 MARKS

D7 Choosing a Savings Account

Difficulty: Reasoning · Amina invests 200 000 CFA for 5 years. Bank X: 4.5% simple. Bank Y: 4% compound annually. Bank Z: 3.8% compounded quarterly.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(d)	Doubling at 4%: $200\,000(1.04)^t = 400\,000 \rightarrow (1.04)^t = 2$ $t = \log 2 / \log 1.04 = 0.30103 / 0.017033 = 17.67$ 17 years gives $(1.04)^{17} = 1.9479$ (not yet double); 18 years gives 2.0258 (doubled). **Answer** **18 complete years** **Marking** *1 mark for the equation · 2 marks for logs · 1 mark for rounding up to a complete year.*	**4**
(e)	Bank Z: quarterly rate 3.8%/4 = 0.95%, with 4 × 5 = 20 quarters. $A = 200\,000(1.0095)^{20} = 200\,000 \times 1.208187 = 241\,637$ CFA Comparison of the three: X 245 000 > Y 243 331 > Z 241 637. Z's nominal 3.8% is actually 3.85% effective, below Y's 4%. **Recommendation:** Y over Z — it pays about 1 690 CFA more after 5 years and is simple to operate. (X pays most over exactly 5 years, but only because its simple rate is higher; beyond about 7 years Y overtakes X.) **Answer** **Z ≈ 241 637 CFA; choose Y over Z (about 1 690 CFA more).** **Marking** *1 mark for the quarterly set-up · 1 mark for the value · 1 mark for a justified recommendation.*	**3**

EXAMINER NOTE *Comparing nominal rates is not enough — convert to effective annual rates. Z's 3.8% quarterly compounding is 3.85% effective, still below Y's 4%.*

CRITERION D CRITERION C

12 MARKS

D8 Study Time vs Test Score

Difficulty: Applying · A scatter plot of study hours against test score for 12 students, with line of best fit y = 2.9x + 7.0.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	**Direction:** positive — more study hours are associated with higher scores. **Strength:** strong — the points lie close to a straight line (correlation roughly 0.9). **Form:** linear; no obvious curve or outliers. **Answer** **Strong positive linear association.** **Marking** *1 mark per feature (direction, strength, form).*	**3**
(b)	Using the printed line y = 2.9x + 7.0: x = 6: y = 2.9(6) + 7.0 = 24.4 x = 11: y = 2.9(11) + 7.0 = 38.9 (If the gradient 3.2 quoted in the task is used instead: 26.2 and 42.2.) **Answer** **≈ 24% for 6 hours and ≈ 39% for 11 hours** **Marking** *1 mark each, with substitution shown.*	**2**

CRITERION D CRITERION C

12 MARKS

D8 Study Time vs Test Score

Difficulty: Applying · A scatter plot of study hours against test score for 12 students, with line of best fit y = 2.9x + 7.0.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(c)	The data cover only about 1 to 10 hours. Predicting at 20 hours extrapolates far beyond the observed range, where the linear pattern may no longer hold. There are also practical limits: nobody can study 20 focused hours a week indefinitely, and the score cannot exceed 100%. **Answer** **Extrapolation beyond the data range (and beyond a realistic study week).** **Marking** *1 mark for extrapolation · 1 mark for the explanation.*	**2**
(d)	Gradient ≈ 2.9 (the task quotes 3.2): each additional hour of study per week is associated with about 3 percentage points more on the test. The intercept ≈ 7 is the predicted score for zero study hours. **Answer** **Each extra hour of weekly study predicts roughly +3 percentage points.** **Marking** *1 mark for the correct direction/magnitude · 1 mark for context and units.*	**2**
(e)	Possible confounders: prior attainment, attendance, the quality of teaching or of revision materials, home environment, motivation, sleep, or test anxiety. Any of these can raise both study time and score, so correlation does not prove that studying causes the higher score. **Answer** **e.g. prior attainment or attendance (affects both variables).** **Marking** *1 mark for a valid variable · 1 mark for explaining the confounding effect · 1 mark for the causal caution.*	**3**

EXAMINER NOTE *Note the internal inconsistency in the task: the plotted line is y = 2.9x + 7.0 while part (d) mentions 3.2. Both readings are accepted; the interpretation is what is marked.*

CRITERION D CRITERION C

14 MARKS

D9 Coastal Navigation

Difficulty: Reasoning · A boat sails from A on bearing 040° for 30 km to B, then on bearing 115° for 40 km to C.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	The bearing of A from B is the reverse bearing: 040° + 180° = 220°. The bearing of C from B is 115°, measured clockwise from north at B. The interior angle between BA and BC is the difference: ∠ABC = 220° − 115° = 105° **Answer** **105° (back-bearing 220° minus bearing 115°)** **Marking** *1 mark for the back-bearing · 1 mark for the difference · 1 mark for the conclusion.*	**3**

CRITERION D | CRITERION C

14 MARKS

D9 Coastal Navigation

Difficulty: Reasoning · A boat sails from A on bearing 040° for 30 km to B, then on bearing 115° for 40 km to C.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	Cosine rule, with AB = 30, BC = 40 and the included angle 105°: $AC^2 = 30^2 + 40^2 - 2(30)(40)\cos 105°$ $= 900 + 1600 - 2400(-0.258819)$ $= 2500 + 621.17 = 3121.17$ $AC = 55.87$ **Answer** AC ≈ 56 km **Marking** *1 mark for the cosine rule · 1 mark for substitution · 1 mark for cos 105° = −0.2588 · 1 mark for the answer.*	**4**
(c)	Use the sine rule to find the angle at A, then add the bearing of AB: $\sin A / 40 = \sin 105° / 55.87 \rightarrow \sin A = 40 \times 0.9659/55.87 = 0.6916$ $A = 43.8°$ Bearing of C from A = 040° + 43.8° = 83.8°. (Coordinate check: C is at (55.5, 6.1) from A, giving atan(55.5/6.1) ≈ 84°.) **Answer** Bearing ≈ 084° **Marking** *1 mark for the sine rule · 1 mark for the angle at A · 1 mark for adding the bearing · 1 mark for rounding.*	**4**
(d)	Time = distance/speed = 55.87/18 = 3.1037 hours. 0.1037 × 60 = 6.2 minutes. **Answer** ≈ 3 hours 6 minutes **Marking** *1 mark for the division · 1 mark for converting the decimal part · 1 mark for the answer in h and min.*	**3**

EXAMINER NOTE *Always sketch and mark north lines at each point — the interior angle 105° comes from a back-bearing, which is where most marks are lost.*

CRITERION D | CRITERION C

15 MARKS

D10 Designing a Cone from a Sector

Difficulty: Extension Challenge · A cone is made from a sector of radius 8 cm and angle 70°.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Arc length = $(\theta/360) \times 2\pi r = (70/360) \times 2\pi(8)$ $= (7/36)(16\pi) = 28\pi/9$ **Answer** $28\pi/9 \approx 9.77$ cm **Marking** *1 mark for the formula · 1 mark for substitution · 1 mark for the exact value.*	**3**

CRITERION D CRITERION C

15 MARKS

D10 Designing a Cone from a Sector

Difficulty: Extension Challenge · A cone is made from a sector of radius 8 cm and angle 70°.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	The arc becomes the circumference of the cone's base: $2\pi r(\text{cone}) = 28\pi/9 \rightarrow r = 14/9$ **Answer** $r = 14/9 \approx 1.56$ cm **Marking** *1 mark for equating to $2\pi r$ · 2 marks for the value.*	**3**
(c)	The sector radius becomes the slant height: $l = 8$. $h^2 + r^2 = l^2 \rightarrow h^2 = 64 - (14/9)^2 = 64 - 2.4198 = 61.5802$ $h = 7.8473\ldots$ **Answer** $h \approx 7.85$ cm **Marking** *1 mark for Pythagoras · 1 mark for substitution · 1 mark for the value.*	**3**
(d)	$V = (1/3)\pi r^2 h = (1/3)\pi(2.4198)(7.8473) = 19.885$ **Answer** $V \approx 19.9$ cm^3 **Marking** *1 mark for the formula · 1 mark for substitution · 1 mark for the answer.*	**3**
(e)	Let the sector angle be θ. Then $r = 8\theta/360$ and $h = \sqrt{(64 - r^2)}$. $V(\theta) = (1/3)\pi r^2 h$. Solve $V(\theta) = 150$ by trial and refinement: $\theta = 200°$: $V \approx 134$ cm^3; $\theta = 215°$: $V \approx 153$ cm^3; $\theta = 212°$: $V \approx 150$ cm^3. Refining gives $\theta \approx 211.7°$. **Answer** $\theta \approx 212°$ **Marking** *1 mark for the volume as a function of θ · 2 marks for a systematic search · 1 mark for $\theta \approx 212°$ (accept 210°–213°).*	**3**

EXAMINER NOTE *Keep exact values ($28\pi/9$, $14/9$) until the last step — rounding early is the main source of error in this type of task.*

SEQUENCES AND PATTERNS

Sequences and Series in Context

5

Two classic models: **arithmetic** (constant difference, linear total) and **geometric** (constant ratio, often with an infinite sum).

CRITERION D | CRITERION B

10 MARKS

S1 Saving Up

Difficulty: Applying · Kofi saves 500 CFA in week 1 and 250 CFA more each week thereafter.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Arithmetic sequence with a = 500 and d = 250: $u_{12} = 500 + 11(250) = 500 + 2750$ **Answer** **3 250 CFA in week 12** **Marking** *1 mark for the rule · 1 mark for the value with units.*	**2**
(b)	Total after n weeks is the sum of an arithmetic series: $T(n) = n/2\,[2a + (n - 1)d]$ $= n/2\,[1000 + 250(n - 1)]$ $= n/2\,[750 + 250n]$ $= 125n(3 + n)$ $T(n) = 125n(n + 3)$ Check n = 3: 500 + 750 + 1000 = 2250 and 125(3)(6) = 2250 ✓ **Answer** **T(n) = 125n(n + 3)** **Marking** *2 marks for the series formula · 1 mark for factorising · 1 mark for the check.*	**4**
(c)	$125n(n + 3) = 70\,000 \rightarrow n(n + 3) = 560$ $n^2 + 3n - 560 = 0$ $n = [-3 + \sqrt{(9 + 2240)}]/2 = (-3 + 47.42)/2 = 22.21$ n must be a whole number: after 22 weeks T = 68 750 CFA, after 23 weeks T = 74 750 CFA. So the target is first passed during week 23. **Answer** **23 weeks (the total passes 70 000 CFA during that week)** **Marking** *1 mark for the equation · 2 marks for solving · 1 mark for interpreting the non-integer result.*	**4**

EXAMINER NOTE *Part (c) is a "first exceeds" question: solve, then round* **up** *and check both sides. Giving n = 22.2 without interpretation loses the final mark.*

CRITERION B CRITERION D

10 MARKS

S2 Bouncing Ball

Difficulty: Reasoning · A ball is dropped from 2 m and rebounds to 80% of its previous height after each bounce.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Each height is 0.8 times the previous one: $h_1 = 2(0.8) = 1.6$ m $h_2 = 1.6(0.8) = 1.28$ m $h_3 = 1.28(0.8) = 1.024$ m **Answer** **1.6 m, 1.28 m, 1.024 m** **Marking** *1 mark per height.*	**3**
(b)	After n bounces the height is the initial height multiplied by 0.8, n times. **Answer** **$h_n = 2(0.8)^n$ metres** **Marking** *1 mark for the structure · 1 mark for the correct formula.*	**2**
(c)	Distance = first fall + 2 × (each rise), because every rebound is travelled up and then down: $D = 2 + 2(1.6 + 1.28 + 1.024 + 0.8192)$ $= 2 + 2(4.7232)$ **Answer** **≈ 11.45 m by the 5th impact** **Marking** *1 mark for the structure (2× rebounds) · 1 mark for the five terms · 1 mark for the total.*	**3**
(d)	As the number of bounces grows, the bounce heights form a geometric series with $\|r\| < 1$, so the total converges: $D_\infty = 2 + 2 \times 1.6/(1 - 0.8) = 2 + 2(8) = 18$ m The model predicts the ball makes infinitely many ever-smaller bounces and comes to rest after travelling about 18 m. In reality it stops when the rebound is too small to see (energy is also lost to sound and heat). **Answer** **The distance converges to 18 m; the ball effectively stops, though the model allows infinitely many bounces.** **Marking** *1 mark for recognising a convergent geometric series · 1 mark for the interpretation.*	**2**

EXAMINER NOTE *The factor of 2 on the rebounds is the most common omission. Only the first fall happens once.*

ALGEBRA

Equations, Inequalities and Fractions

6

Accuracy of manipulation and clarity of method: every line should follow logically from the one above it, and every solution should be checked.

CRITERION C

12 MARKS

A1 Solving Equations — Mixed Practice

Difficulty: Applying · Solve each equation or inequality, showing all steps.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	$3(2x - 5) = 4(x + 1) - 7$ $6x - 15 = 4x + 4 - 7$ $6x - 15 = 4x - 3$ $2x = 12$ $x = 6$ Check: $3(12 - 5) = 21$ and $4(7) - 7 = 21$ ✓ **Answer** $x = 6$ **Marking** *1 mark for expanding both sides · 1 mark for collecting terms · 1 mark for x = 6.*	**3**
(b)	$2x + y = 7$ (1) $3x - 2y = 8$ (2) From (1): $y = 7 - 2x$. Substitute into (2): $3x - 2(7 - 2x) = 8$ $3x - 14 + 4x = 8$ $7x = 22 \rightarrow x = 22/7$ $y = 7 - 2(22/7) = 49/7 - 44/7 = 5/7$ Check in (2): $3(22/7) - 2(5/7) = 66/7 - 10/7 = 56/7 = 8$ ✓ **Answer** $x = 22/7 \approx 3.14, y = 5/7 \approx 0.71$ **Marking** *1 mark for substitution · 1 mark for solving x · 1 mark for y (exact fractions preferred).*	**3**
(c)	$5 - 2x \leq 13$ $-2x \leq 8$ Divide by −2 and **reverse the inequality**: $x \geq -4$ Number line: a solid circle at −4 with the arrow pointing right. **Answer** $x \geq -4$ **Marking** *1 mark for rearranging · 1 mark for reversing the sign · 1 mark for a correct number line.*	**3**

CRITERION C

12 MARKS

A1 Solving Equations — Mixed Practice

Difficulty: Applying · Solve each equation or inequality, showing all steps.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(d)	$s = ut + \frac{1}{2}at^2 \rightarrow \frac{1}{2}at^2 + ut - s = 0$ Multiply by 2: $at^2 + 2ut - 2s = 0$ This is a quadratic in t with A = a, B = 2u, C = −2s, so $t = [-2u \pm \sqrt{(4u^2 + 8as)}]/(2a) = [-u \pm \sqrt{(u^2 + 2as)}]/a$ **Answer** $at^2 + 2ut - 2s = 0$, giving $t = [-u \pm \sqrt{(u^2 + 2as)}]/a$ **Marking** *1 mark for rearranging · 1 mark for identifying the quadratic · 1 mark for the formula.*	**3**

EXAMINER NOTE *In (d) the equation is ready to solve but not solved for a specific number: leaving it in general form with the quadratic formula is the expected answer.*

CRITERION C

12 MARKS

A2 Algebraic Fractions

Difficulty: Reasoning · Simplify, combine and solve expressions involving algebraic fractions.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Factorise both parts: $x^2 - 4x - 5 = (x - 5)(x + 1)$ $x^2 - 1 = (x - 1)(x + 1)$ $(x - 5)(x + 1)/[(x - 1)(x + 1)] = (x - 5)/(x - 1), x \neq \pm 1$ **Answer** $(x - 5)/(x - 1), x \neq 1, x \neq -1$ **Marking** *1 mark per factorisation · 1 mark for cancelling · 1 mark for the restrictions.*	**4**
(b)	Common denominator (x + 1)(x − 2): $2/(x + 1) + 3/(x - 2) = [2(x - 2) + 3(x + 1)]/[(x + 1)(x - 2)]$ $= (2x - 4 + 3x + 3)/[(x + 1)(x - 2)]$ $= (5x - 1)/[(x + 1)(x - 2)]$ **Answer** $(5x - 1)/[(x + 1)(x - 2)], x \neq -1, 2$ **Marking** *1 mark for the common denominator · 2 marks for the numerator · 1 mark for the restrictions.*	**4**

CRITERION C

12 MARKS

A2 Algebraic Fractions

Difficulty: Reasoning · Simplify, combine and solve expressions involving algebraic fractions.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(c)	$3/x + 2/(x + 1) = 4, x \neq 0, -1$ Multiply through by x(x + 1): $3(x + 1) + 2x = 4x(x + 1)$ $3x + 3 + 2x = 4x^2 + 4x$	**4**

CRITERION C

12 MARKS

A2 Algebraic Fractions

Difficulty: Reasoning · Simplify, combine and solve expressions involving algebraic fractions.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

$4x^2 - x - 3 = 0$
$(4x + 3)(x - 1) = 0$
$x = 1$ or $x = -3/4$
Checks: $x = 1 \rightarrow 3 + 1 = 4$ ✓; $x = -0.75 \rightarrow -4 + 8 = 4$ ✓

Answer **$x = 1$ or $x = -3/4$**

Marking *1 mark for clearing the fractions · 1 mark for the quadratic · 1 mark for both roots · 1 mark for checking.*

EXAMINER NOTE *Restrictions belong with the answer, not just the working — state them wherever a denominator could be zero.*

QUADRATIC RELATIONSHIPS

Modelling with Quadratics

7

Quadratic models describe projectile motion. The vertex gives the maximum, the roots give the times when the object is at ground level, and the axis of symmetry links the two.

CRITERION D CRITERION C

14 MARKS

Q1 Projectile Motion

Difficulty: Reasoning · A ball is thrown from a 2 m platform: $h(t) = -5t^2 + 15t + 2$, height in metres, time in seconds.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(a)

At $t = 0$: $h(0) = -5(0) + 15(0) + 2 = 2$.
The constant term of the model is exactly the initial height.

Answer **2 m — the constant term in the formula.**

Marking *1 mark for the value · 1 mark for linking it to the constant term.*

2

(b)

The ball hits the ground when $h = 0$:
$-5t^2 + 15t + 2 = 0 \rightarrow 5t^2 - 15t - 2 = 0$
$t = [15 \pm \sqrt{(225 + 40)}]/10 = [15 \pm \sqrt{265}]/10$
$t = (15 + 16.279)/10 = 3.128$ or $t = (15 - 16.279)/10 = -0.128$
The negative root is rejected (time cannot be negative before the throw).

Answer **$t \approx 3.13$ s**

Marking *1 mark for setting $h = 0$ · 1 mark for the quadratic formula · 1 mark for the positive root · 1 mark for rejecting the negative root.*

4

CRITERION D | CRITERION C

14 MARKS

Q1 Projectile Motion

Difficulty: Reasoning · A ball is thrown from a 2 m platform: $h(t) = -5t^2 + 15t + 2$, height in metres, time in seconds.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(c)	Complete the square: $h = -5(t^2 - 3t) + 2 = -5[(t - 1.5)^2 - 2.25] + 2$ $= -5(t - 1.5)^2 + 11.25 + 2$ $= -5(t - 1.5)^2 + 13.25$ So the maximum height is 13.25 m, reached at $t = 1.5$ s. Check by symmetry: the vertex is midway between the roots, $t = (3.128 - 0.128)/2 = 1.5$ ✓ **Answer** Maximum 13.25 m at $t = 1.5$ s **Marking** *2 marks for completing the square · 1 mark for the maximum height · 1 mark for the time.*	**4**
(d)	Set $h = 8$: $-5t^2 + 15t + 2 = 8 \rightarrow 5t^2 - 15t + 6 = 0$ $t = [15 \pm \sqrt{(225 - 120)}]/10 = [15 \pm \sqrt{105}]/10 = [15 \pm 10.247]/10$ $t = 0.475$ s and $t = 2.525$ s Interpretation: the ball passes 8 m on the way up (0.48 s) and again on the way down (2.52 s). **Answer** $t \approx 0.48$ s (rising) and $t \approx 2.52$ s (falling) **Marking** *1 mark for the equation · 2 marks for both roots · 1 mark for the interpretation.*	**4**

EXAMINER NOTE *Both roots of (d) are physically meaningful, and the two answers are symmetric about the vertex at $t = 1.5$ s — a good way to check.*

FUNCTIONS AND GRAPHS

Trigonometric Models

8

Sine and cosine models describe anything that repeats: tides, daylight, temperatures, circuits. Identify the midline, the amplitude and the period before calculating.

CRITERION D | CRITERION B

14 MARKS

F1 Tidal Heights

Difficulty: Extension Challenge · Harbour depth $d(t) = 4 + 2\sin(30t°)$ metres, t hours after midnight.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	At $t = 0$: $d(0) = 4 + 2\sin 0° = 4 + 0$. **Answer** 4 m **Marking** *1 mark for substitution · 1 mark for the answer.*	**2**

CRITERION D | CRITERION B

14 MARKS

F1 Tidal Heights

Difficulty: Extension Challenge · Harbour depth d(t) = 4 + 2 sin(30t°) metres, t hours after midnight.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	The sine term ranges from −1 to +1, so d ranges from 4 − 2 = 2 m to 4 + 2 = 6 m. Maximum when sin(30t°) = 1: 30t = 90 → t = 3 h (and every 12 hours after: 15 h, 27 h…). Minimum when sin(30t°) = −1: 30t = 270 → t = 9 h (and 21 h…). **Answer** **Max 6 m at t = 3 h; min 2 m at t = 9 h.** **Marking** *1 mark for the range · 1 mark for the maximum time · 1 mark for the minimum time · 1 mark for the values.*	**4**
(c)	The coefficient of t is 30 degrees per hour, so a full 360° cycle takes T = 360/30 = 12 hours. **Answer** **12 hours (semidiurnal tide)** **Marking** *1 mark for the method · 1 mark for 12 h.*	**2**
(d)	4 + 2 sin(30t°) ≥ 3 → sin(30t°) ≥ −0.5 In the first 12 hours, 30t runs from 0° to 360°. Since sin x ≥ −0.5 for 0° ≤ x ≤ 210° and 330° ≤ x ≤ 360°: 30t ≤ 210 → t ≤ 7 h and 30t ≥ 330 → t ≥ 11 h So the depth is at least 3 m from midnight to 7 h, and again from 11 h to noon: 8 hours in total. **Answer** **0 ≤ t ≤ 7 and 11 ≤ t ≤ 12 — 8 hours in total.** **Marking** *2 marks for the correct inequality in sine · 1 mark for the intervals · 1 mark for the total.*	**4**
(e)	The model assumes a perfectly regular, symmetric tide with a constant period and amplitude. Real tides vary with the moon's phase, the season, wind and atmospheric pressure, and many coasts have two unequal high tides per day. **Answer** **Real tides are not perfectly sinusoidal or constant in amplitude.** **Marking** *1 mark for a valid limitation · 1 mark for the explanation.*	**2**

EXAMINER NOTE *The 12-hour period means the pattern repeats at t + 12. Answers that list only the first occurrence of a maximum or minimum are incomplete.*

GEOMETRY AND MEASUREMENT

Circle Theorems, Solids and Scale

9

Diagrams, theorems and units: state the theorem you use, keep exact values where possible, and give answers with correct units.

CRITERION C

10 MARKS

G1 Intersecting Chords

Difficulty: Applying · Chords AB and CD intersect at E inside a circle.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	If two chords intersect inside a circle, the products of the segments of each chord are equal: AE × EB = CE × ED **Answer** **AE · EB = CE · ED** **Marking** *2 marks for the complete statement.*	**2**
(b)	6 × 4 = 3 × ED 24 = 3 ED ED = 8 **Answer** **ED = 8** **Marking** *1 mark for the equation · 1 mark for solving · 1 mark for units/statement.*	**3**
(c)	CD = CE + ED = 3 + 8 **Answer** **CD = 11** **Marking** *1 mark for the sum · 1 mark for the answer.*	**2**
(d)	AB is a diameter of length 12, so the radius is 6 and the centre is 6 cm from each end. E is 2 cm from the centre, so AE = 6 + 2 = 8 and EB = 6 − 2 = 4 AE × EB = 8 × 4 = 32, so CE × ED = 32. If CD is perpendicular to AB, then E is the midpoint of CD: CE = ED = √32 = 4√2. CD = 2 × 4√2 = 8√2 ≈ 11.3 cm. **Answer** **CE · ED = 32; if CD ⊥ AB then CD = 8√2 ≈ 11.3 cm.** **Marking** *1 mark for splitting the diameter · 1 mark for the product 32 · 1 mark for the conclusion.*	**3**

EXAMINER NOTE *Part (d) is deliberately open: the product relationship always holds, but the length CD can only be found if CD is also perpendicular to the diameter — which is what makes E the midpoint.*

CRITERION D CRITERION C

13 MARKS

G2 Volumes of 3D Shapes

Difficulty: Applying · A cone with r = 5 cm, h = 18 cm and a sphere with r = 6 cm.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	$V(\text{cone}) = (1/3)\pi r^2 h = (1/3)\pi(25)(18)$ $= 150\pi$ **Answer** **$150\pi \approx 471.2\ \text{cm}^3$** **Marking** *1 mark for the formula · 1 mark for substitution · 1 mark for the exact value.*	**3**
(b)	$V(\text{sphere}) = (4/3)\pi r^3 = (4/3)\pi(216)$ $= 288\pi$ **Answer** **$288\pi \approx 904.8\ \text{cm}^3$** **Marking** *1 mark for the formula · 1 mark for substitution · 1 mark for the exact value.*	**3**
(c)	$288\pi - 150\pi = 138\pi \approx 433.5\ \text{cm}^3$ **Answer** **The sphere is larger by $138\pi \approx 433.5\ \text{cm}^3$.** **Marking** *1 mark for the comparison · 1 mark for the difference.*	**2**
(d)	Removing the sphere leaves a cylindrical hole of the same volume. $V = \pi r^2 \Delta h \rightarrow 288\pi = \pi(6)^2 \Delta h = 36\pi\ \Delta h$ $\Delta h = 288\pi/36\pi = 8$ **Answer** **The level drops by 8 cm.** **Marking** *1 mark for equating volumes · 1 mark for dividing by 36π · 1 mark for the answer.*	**3**
(e)	Volume is conserved when the cone is recast: $150\pi = \pi(5)^2 h \rightarrow 150\pi = 25\pi\ h \rightarrow h = 6$ **Answer** **h = 6 cm** **Marking** *1 mark for equating volumes · 1 mark for the answer.*	**2**

EXAMINER NOTE *Leaving answers in terms of π first makes (c) and (d) almost trivial — exact values are also easier to check.*

CRITERION D CRITERION C

12 MARKS

G3 Pyramid Roof

Difficulty: Reasoning · A square base of side 6 m with a pyramid roof whose apex is 4 m above the centre.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	The slant height is the hypotenuse of a right triangle whose legs are the height (4 m) and the distance from the centre to the midpoint of a side (3 m): $l^2 = 4^2 + 3^2 = 16 + 9 = 25$ $l = 5$ **Answer** **Slant height = 5 m** **Marking** *1 mark for identifying the triangle · 1 mark for Pythagoras · 1 mark for the answer.*	**3**

CRITERION D CRITERION C

12 MARKS

G3 Pyramid Roof

Difficulty: Reasoning · A square base of side 6 m with a pyramid roof whose apex is 4 m above the centre.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	Each face is a triangle with base 6 m and slant height 5 m: Area of one face = ½ × 6 × 5 = 15 m² Four faces = 4 × 15 = 60 m² **Answer** **15 m² per face; 60 m² in total** **Marking** *1 mark for the triangle area formula · 1 mark for one face · 1 mark for the total · 1 mark for units.*	**4**
(c)	60 m² × 7500 CFA/m² **Answer** **450 000 CFA** **Marking** *1 mark for the multiplication · 1 mark for the answer with units.*	**2**
(d)	V = (1/3) × base area × height = (1/3) × (6 × 6) × 4 = (1/3) × 36 × 4 **Answer** **48 m³** **Marking** *1 mark for the formula · 1 mark for the base area · 1 mark for the volume.*	**3**

EXAMINER NOTE *The slant height is not the height of the pyramid: the triangle uses half the base (3 m), not the full side (6 m).*

STATISTICS

Working with Grouped Data

10

For grouped data, estimate each value by the class mid-point. The shape of the distribution tells you how mean, median and mode will compare.

CRITERION D CRITERION C

12 MARKS

St1 Analysing Height Data

Difficulty: Applying · Heights of 200 students in 5 cm classes from 140–145 to 180–185.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Frequencies rise to a peak in 155–160 (44 students) and then fall away on both sides, roughly symmetrically. The modal class is the class with the highest frequency. **Answer** **Roughly symmetric, single-peaked; modal class 155–160 cm.** **Marking** *1 mark for shape · 1 mark for the modal class · 1 mark for using frequencies.*	**3**

CRITERION D | CRITERION C

12 MARKS

St1 Analysing Height Data

Difficulty: Applying · Heights of 200 students in 5 cm classes from 140–145 to 180–185.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(b) Use class mid-points × frequency, then divide by 200. **5**

Class (cm)	Mid-point	Frequency	Mid-point × frequency
140–145	142.5	8	1 140
145–150	147.5	18	2 655
150–155	152.5	32	4 880
155–160	157.5	44	6 930
160–165	162.5	42	6 825
165–170	167.5	29	4 857.5
170–175	172.5	17	2 932.5
175–180	177.5	8	1 420
180–185	182.5	2	365
Total	—	200	32 005

Answer **Estimated mean = 32 005/200 = 160.0 cm**

Marking *2 marks for mid-points · 1 mark for the products · 1 mark for the total · 1 mark for the mean.*

(c) The distribution is close to symmetric, so the balance point (mean) and the middle value (median) nearly coincide. **2**

Skewness pulls the mean towards the longer tail; here there is very little skew (only 2 students above 180 and 26 below 150).

Answer **The data are nearly symmetric, so mean ≈ median.**

Marking *1 mark for "symmetric" · 1 mark for explaining the effect of skew.*

(d) Students between 150 and 175 cm: 32 + 44 + 42 + 29 + 17 = 164. **2**

164/200 = 82%.

So 82% — about four in five — lie in that range, but 36 students (18%) lie outside it.

Answer **Broadly true but overstated: 82%, not "almost all".**

Marking *1 mark for the calculation · 1 mark for evaluating the wording.*

EXAMINER NOTE *"Almost all" is an evaluative phrase: quote the percentage and say whether the phrase is justified. 82% is a large majority, not almost all.*

PROBABILITY

Tree Diagrams Without Replacement 11

Without replacement, the denominator of the second branch changes, and so do the numerators. Every path probability is a product; paths that satisfy the event are added.

CRITERION D CRITERION C

12 MARKS

P1 Two Marbles Without Replacement

Difficulty: Applying · A bag contains 5 red and 4 blue marbles; two are drawn without replacement.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(a) First draw: 9 marbles in total, 5 red and 4 blue.

Second draw: only 8 marbles remain, and one marble of the first colour has gone.

Note: the tree printed in the question paper shows 3/5 and 2/5 on the first branches and 1/2, 3/4 on the second — those values do not match 5 red and 4 blue. The correct tree is:

4

First draw	Second draw	Path	Probability
Red 5/9	Red 4/8	RR	20/72 = 5/18
Red 5/9	Blue 4/8	RB	20/72 = 5/18
Blue 4/9	Red 5/8	BR	20/72 = 5/18
Blue 4/9	Blue 3/8	BB	12/72 = 1/6

Answer **P(R) = 5/9, P(B) = 4/9; then 4/8 or 5/8, and 4/8 or 3/8.**

Marking *1 mark for the first branches · 2 marks for the second branches · 1 mark for correct path products.*

(b) P(RR) = (5/9) × (4/8) = 20/72

2

Answer **5/18 ≈ 0.278**

Marking *1 mark for the product · 1 mark for the simplified fraction.*

(c) Different colours means RB or BR:

P(RB) = (5/9)(4/8) = 5/18

P(BR) = (4/9)(5/8) = 5/18

P(different) = 5/18 + 5/18 = 10/18

3

Answer **5/9 ≈ 0.556**

Marking *1 mark for the two paths · 1 mark for adding · 1 mark for simplifying.*

CRITERION D CRITERION C

12 MARKS

P1 Two Marbles Without Replacement

Difficulty: Applying · A bag contains 5 red and 4 blue marbles; two are drawn without replacement.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(d) P(second red) = P(RR) + P(BR) = 5/18 + 5/18 = 10/18 = 5/9.

3

CRITERION D CRITERION C

12 MARKS

P1 Two Marbles Without Replacement

Difficulty: Applying · A bag contains 5 red and 4 blue marbles; two are drawn without replacement.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

This equals P(first red) = 5/9.

Comment: the second draw is not affected in the long run — the marbles are exchangeable, so any draw has the same 5/9 chance of being red. Removing a marble changes the second probability *given* the first result, but not the overall probability.

Answer **5/9 — the same as the first draw.**

Marking *1 mark for the calculation · 2 marks for the insight about symmetry/exchangeability.*

EXAMINER NOTE *Part (d) is the classic "surprising symmetry" result. The check is that the total probability of red on the second draw must be 5/9, since the colours are randomly arranged in the bag.*

FINANCIAL MATHEMATICS

Depreciation and Reducing Balance 12

Reducing-balance depreciation is compound interest with a negative rate: multiply by (1 − r) each year. Straight-line depreciation subtracts a fixed amount instead.

CRITERION D CRITERION B

12 MARKS

Fn1 Depreciating a Motorcycle

Difficulty: Applying · A new motorcycle costs 1 200 000 CFA and depreciates at 12% per year (reducing balance).

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS

(a)

After 1 year: $1\,200\,000(0.88) = 1\,056\,000$

After 3 years: $1\,200\,000(0.88)^3 = 1\,200\,000 \times 0.681472 = 817\,766.4$

Answer **1 056 000 CFA after 1 year; ≈ 817 766 CFA after 3 years**

Marking *1 mark for the multiplier 0.88 · 1 mark for year 1 · 1 mark for year 3.*

3

(b)

Each year the value is 88% of the previous year's value.

Answer **$V(t) = 1\,200\,000(0.88)^t$**

Marking *1 mark for the structure · 1 mark for the correct constants.*

2

CRITERION D | CRITERION B

12 MARKS

Fn1 Depreciating a Motorcycle

Difficulty: Applying · A new motorcycle costs 1 200 000 CFA and depreciates at 12% per year (reducing balance).

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(c)	$1\,200\,000(0.88)^t < 500\,000 \rightarrow (0.88)^t < 0.416667$ $t > \log(0.416667)/\log(0.88) = (-0.38021)/(-0.05552) = 6.85$ So the value is still above 500 000 after 6 years (557 285 CFA) and below it after 7 years (490 411 CFA). **Answer** **After 7 complete years** **Marking** *1 mark for the inequality · 2 marks for solving with logs · 1 mark for the year and check.*	**4**
(d)	Reducing balance after 5 years: $1\,200\,000(0.88)^5 = 1\,200\,000 \times 0.527732 = 633\,278$ CFA Straight line: 12% of 1 200 000 = 144 000 per year, so V = 1 200 000 − 5(144 000) = 480 000 CFA Difference: 633 278 − 480 000 = 153 278 CFA **Answer** **Reducing balance is higher by ≈ 153 278 CFA.** **Marking** *1 mark for the reducing-balance value · 1 mark for the straight-line value · 1 mark for the comparison.*	**3**

EXAMINER NOTE *Reducing balance depreciates more slowly in money terms at first, so it leaves a higher value than straight line for the same nominal rate.*

MIXED CHALLENGE

MYP 5 eAssessment Challenges

13

These tasks combine topics in unfamiliar settings. Before calculating, write down the model you will use and the assumptions it makes.

CRITERION D | CRITERION C | CRITERION B

18 MARKS

M1 Planning a Concert

Difficulty: eAssessment Challenge · A hall seats 800. Fixed costs: venue 500 000, performer 300 000, security 200 000 CFA; variable cost 500 CFA per person. Ticket price x CFA. Demand N(x) = 800 − 0.05x.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	Fixed costs = 500 000 + 300 000 + 200 000 = 1 000 000 CFA Variable cost for 800 tickets = 800 × 500 = 400 000 CFA Revenue if all 800 sell at x CFA = 800x **Answer** **1 000 000 CFA; 400 000 CFA; 800x CFA** **Marking** *1 mark for fixed · 1 mark for variable · 2 marks for revenue.*	**4**

CRITERION D | CRITERION C | CRITERION B

18 MARKS

M1 Planning a Concert

Difficulty: eAssessment Challenge · A hall seats 800. Fixed costs: venue 500 000, performer 300 000, security 200 000 CFA; variable cost 500 CFA per person. Ticket price x CFA. Demand N(x) = 800 − 0.05x.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(b)	Profit = revenue − total cost P(x) = 800x − (1 000 000 + 400 000) P(x) = 800x − 1 400 000 **Answer** **P(x) = 800x − 1 400 000** **Marking** *2 marks for the cost structure · 1 mark for the simplified form.*	**3**
(c)	Break-even: 800x − 1 400 000 = 0 x = 1 400 000/800 = 1750 **Answer** **1 750 CFA per ticket** **Marking** *1 mark for setting P = 0 · 1 mark for the value.*	**2**
(d)	A constant 800 tickets sold at every price is unrealistic: as the price rises, some people decide not to come. The model N(x) = 800 − 0.05x reduces the number sold by 0.05 tickets for every 1 CFA increase (50 tickets for each 1000 CFA), which matches the usual downward-sloping demand. It also stays sensible: at x = 0 (free entry) N = 800, the full capacity. **Answer** **Demand falls as price rises; it is capped at the 800-seat capacity.** **Marking** *2 marks for the reasoning · 1 mark for linking to capacity/context.*	**3**
(e)	Revenue = price × number sold = x(800 − 0.05x) Variable cost = 500 × N(x) = 500(800 − 0.05x) = 400 000 − 25x Π(x) = x(800 − 0.05x) − 1 000 000 − (400 000 − 25x) = 800x − 0.05x² − 1 000 000 − 400 000 + 25x Π(x) = −0.05x² + 825x − 1 400 000 **Answer** **Π(x) = −0.05x² + 825x − 1 400 000** **Marking** *1 mark for revenue · 1 mark for variable cost · 1 mark for simplifying.*	**3**
(f)	The quadratic opens downwards, so the vertex gives the maximum. x = −b/(2a) = −825/(2 × −0.05) = 8250 Π(8250) = −0.05(8250)² + 825(8250) − 1 400 000 = −3 403 125 + 6 806 250 − 1 400 000 = 2 003 125 Tickets sold: N(8250) = 800 − 412.5 = 387.5 ≈ 388. So the optimum is a high price with only about half the hall full. **Answer** **x = 8 250 CFA, maximum profit ≈ 2 003 125 CFA (≈ 388 tickets).** **Marking** *1 mark for the vertex method · 1 mark for the price · 1 mark for the profit.*	**3**

EXAMINER NOTE *The "all tickets sell" model and the demand model give very different answers — a good discussion point about how modelling assumptions change conclusions.*

CRITERION D | CRITERION B | CRITERION C

18 MARKS

M2 Cell Growth and Medication

Difficulty: eAssessment Challenge · Strain P: P(t) = 200 + 80t. Strain Q: Q(t) = 50 × 2^(t/3). Medication kills 90% of Q when Q exceeds 5000.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	$P(0) = 200 + 0 = 200$ $Q(0) = 50 \times 2^0 = 50$ **Answer** **P: 200; Q: 50** **Marking** *1 mark each.*	**2**
(b)	At t = 0 the linear strain grows at 80 cells per hour. Q grows exponentially; its instantaneous rate at t = 0 is $Q'(0) = 50 \times (\ln 2)/3 \approx 11.5$ cells per hour. So P is growing faster at t = 0 — but Q's rate doubles every 3 hours. **Answer** **P (80 per hour vs ≈ 11.5 per hour initially).** **Marking** *1 mark for P's rate · 1 mark for comparing with Q · 1 mark for the conclusion.*	**3**
(c)	t = 6: $P = 200 + 480 = 680$; $Q = 50 \times 2^2 = 200$ t = 12: $P = 200 + 960 = 1160$; $Q = 50 \times 2^4 = 800$ **Answer** **t = 6: 680 and 200; t = 12: 1 160 and 800** **Marking** *1 mark per correct pair (2) · 2 marks for accuracy.*	**4**
(d)	$50 \times 2^{(t/3)} = 5000 \rightarrow 2^{(t/3)} = 100$ $t/3 = \log_2 100 = 6.644$ $t = 19.93$ **Answer** **t ≈ 20 hours** **Marking** *1 mark for dividing by 50 · 2 marks for logs · 1 mark for rounding to the nearest hour.*	**3**
(e)	Just before treatment Q ≈ 5080, so after killing 90% about 508 remain. From then on $Q(t) = 508 \times 2^{((t-20)/3)}$. Q exceeds 5000 again when $2^{((t-20)/3)} = 10 \rightarrow (t-20)/3 = 3.322$ $t - 20 = 9.97 \rightarrow t \approx 30$ hours **Answer** **About 10 hours later, at t ≈ 30 hours.** **Marking** *1 mark for the reduced population · 1 mark for the factor of 10 · 1 mark for the time.*	**3**
(f)	In the long run the exponential strain dominates: Q grows without limit while P grows only linearly, so Q becomes vastly larger than P. Limitations: unlimited exponential growth is unrealistic (nutrients run out), the "instant" 90% kill ignores how the drug is absorbed, and the model has no resistance or repeated dosing. **Answer** **Q dominates long-term; both models ignore resource limits and drug dynamics.** **Marking** *1 mark for the long-term comparison · 2 marks for limitations.*	**3**

EXAMINER NOTE *Part (e) is a two-stage exponential: reset the population, then restart the clock. Students often answer 20 hours again.*

CRITERION D | CRITERION C

16 MARKS

M3 The Clock Tower

Difficulty: eAssessment Challenge · From C (25 m from the base B) the angle of elevation to the top A is 32°; from D, further back on the same line, it is 20°.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	A labelled diagram should show: a vertical tower AB with B at ground level, C and D on the horizontal line through B, and the two right angles at B. Mark 25 m from B to C, the angle 32° at C and 20° at D, with D beyond C. **Answer** **Vertical tower, horizontal ground, both angles measured from the ground at C and D.** **Marking** *1 mark for the layout · 1 mark for the labels.*	**2**
(b)	In the right triangle ABC, AB is opposite the 32° angle and BC = 25 is adjacent: $\tan 32° = AB/25 \rightarrow AB = 25 \tan 32° = 25 \times 0.624869$ **Answer** **AB ≈ 15.6 m** **Marking** *1 mark for the tangent ratio · 1 mark for substitution · 1 mark for the answer to 0.1 m.*	**3**
(c)	For the larger triangle ABD: $\tan 20° = AB/BD \rightarrow BD = AB/\tan 20°$ BD = 15.6217/0.363970 = 42.920 CD = BD − BC = 42.920 − 25 **Answer** **BD ≈ 42.9 m; CD ≈ 17.9 m** **Marking** *2 marks for BD · 1 mark for the subtraction · 1 mark for the answer.*	**4**
(d)	The clock has diameter 2 m centred at height AB, so its top and bottom are at heights AB + 1 and AB − 1: top: $\tan \alpha = 16.6217/25 \rightarrow \alpha = 33.62°$ bottom: $\tan \beta = 14.6217/25 \rightarrow \beta = 30.32°$ Angle subtended = $\alpha - \beta = 3.30°$ **Answer** **≈ 3.3° (between the lines of sight to the top and bottom of the clock)** **Marking** *2 marks for the two heights · 1 mark for both angles · 1 mark for the difference.*	**4**
(e)	**Assumption:** the ground is horizontal and level, the tower is vertical, and C, D, B are collinear. **Error source:** the clinometer reading of 1°–2°, the measurement of 25 m, or the height of the observer's eye if the angles were taken from eye level rather than the ground. **Answer** **Level ground/vertical tower; angle and distance measurement error.** **Marking** *1 mark for an assumption · 1 mark for an error source · 1 mark for explanation.*	**3**

EXAMINER NOTE *Part (d) is an angle subtended question: find the elevation of the top and of the bottom, then subtract. Do not simply halve the diameter.*

CRITERION B | CRITERION C | CRITERION D

14 MARKS

M4 Rational Function Drug Model

Difficulty: eAssessment Challenge · Drug concentration $C(t) = 20t/(t^2 + 4)$ mg/L, $t \geq 0$ hours after injection.

PART	WORKED SOLUTION AND MARKING GUIDE	MARKS
(a)	$C(1) = 20(1)/(1 + 4) = 20/5$ $C(4) = 80/(16 + 4) = 80/20$ **Answer** **C(1) = 4 mg/L and C(4) = 4 mg/L** **Marking** *1 mark each.*	**2**
(b)	The degree of the denominator (2) exceeds the degree of the numerator (1), so $C(t) \to 0$ as $t \to \infty$. Horizontal asymptote: $y = 0$ (the t-axis). Meaning: the drug is eventually eliminated, so the concentration approaches zero. **Answer** **y = 0; concentration tends to zero in the long run.** **Marking** *1 mark for the asymptote · 2 marks for the meaning/reasoning.*	**3**
(c)	$20t/(t^2 + 4) = 4 \to 20t = 4t^2 + 16$ $4t^2 - 20t + 16 = 0 \to t^2 - 5t + 4 = 0$ $(t - 1)(t - 4) = 0$ $t = 1$ or $t = 4$ **Answer** **t = 1.00 h and t = 4.00 h** **Marking** *1 mark for clearing the fraction · 1 mark for the quadratic · 2 marks for both roots.*	**4**
(d)	Complete the square in the denominator, or use the quotient rule: $C'(t) = [20(t^2 + 4) - 20t(2t)]/(t^2 + 4)^2 = (80 - 20t^2)/(t^2 + 4)^2$ Set $C' = 0$: $80 - 20t^2 = 0 \to t^2 = 4 \to t = 2$ $(t \geq 0)$. $C(2) = 40/8 = 5$ So the maximum concentration is 5 mg/L at $t = 2$ hours. **Answer** **Maximum 5 mg/L at t = 2 h** **Marking** *1 mark for the derivative/approach · 1 mark for t = 2 · 1 mark for 5 mg/L.*	**3**
(e)	$C(t) \geq 4$ between the two solutions found in (c): from $t = 1$ to $t = 4$. Duration = 4 − 1 = 3 hours. **Answer** **3 hours ($1 \leq t \leq 4$)** **Marking** *1 mark for the interval · 1 mark for the duration.*	**2**

EXAMINER NOTE *Parts (c)–(e) connect: the two solutions of C = 4 are the times the drug enters and leaves the effective range, and the peak sits midway at t = 2.*

MARK SCHEME

How the Marks Are Awarded

A

The conventions used throughout this answer sheet, and how they map to the three MYP criteria.

M — method marks are for a valid approach: choosing the correct formula, setting up an equation, drawing a correct diagram, or starting a valid chain of reasoning. A method mark can be earned even if the arithmetic later slips.

A — accuracy marks are for correct values, correct units and correct rounding. They usually depend on the method mark in the same part.

R — reasoning and communication marks are for explanation, justification, correct notation and a logical sequence of steps. This is where Criterion C is assessed.

D — interpretation marks are for reading the answer back into the context: stating what it means, checking whether it is reasonable, naming assumptions and limitations.

Follow-through: an arithmetic slip early in a question does not automatically lose later marks — a correct method applied to an incorrect value still earns method marks. Accuracy marks, however, are lost.

Equivalent methods are always accepted: trial and improvement, graphing, the quadratic formula or completing the square are all valid where the mathematics allows. This document shows *a* model solution, not *the only* solution.

Rounding: keep full accuracy in the working and round only at the end, to the precision requested (or 3 significant figures where none is stated). Premature rounding is the most common cause of lost accuracy marks.

Criterion	What is assessed	What earns the top band
B — Investigating patterns	Recognising patterns, describing them as relationships or general rules, and justifying or proving them.	A general rule stated algebraically, tested on further cases, and justified structurally — not just inferred from a table.
C — Communicating	Using forms of representation, moving between them, and communicating reasoning with correct notation and logical structure.	Clear, sequenced working; correct notation and units; labelled diagrams; every step justified.
D — Applying mathematics	Identifying relevant elements, selecting appropriate tools, applying them and evaluating the results in context.	An explicit model, stated assumptions, accurate calculation, and a critical evaluation of reasonableness and limitations.

Each criterion is marked out of 8, so the four criteria together give a maximum of 32 — the structure used by the MYP Mathematics eAssessment. The bands below describe the kind of performance associated with each level.

Level	Criterion B	Criterion C	Criterion D
1–2	Makes a limited attempt to use a strategy; recognises little or no pattern.	Uses limited forms of representation; communication is unclear or incomplete.	Identifies few relevant elements; applies little mathematics.
3–4	Applies a strategy with guidance; describes a simple pattern.	Uses appropriate forms of representation, with some success in communicating reasoning.	Identifies some relevant elements; applies straightforward tools with partial success.
5–6	Selects and applies appropriate strategies; describes patterns as general rules and begins to justify them.	Moves between forms of representation; communicates coherent reasoning with generally correct notation.	Selects appropriate tools; applies them accurately; gives a reasonable interpretation of results.
7–8	Uses flexible and efficient strategies; states general rules and justifies or proves them, testing and refining.	Moves fluently between forms; communicates complete, precise and logically structured reasoning.	Identifies relevant elements fully, models effectively, evaluates results critically and discusses limitations.

These descriptors are a study aid summarising the spirit of the criteria, not an official IB publication. Always use your school's task-specific clarifications when a task is actually assessed.

Every task in the bank carries one criterion as its **main** focus and often one or two supporting criteria. Use the table below to build a revision plan that targets a specific criterion.

Criterion	Main criterion	Supporting criterion	Marks
B — Investigating patterns	B1, B2, B3, B4, B5, B6, S2, M4	C3, C4, C7, D2, D7, S1, F1, Fn1, M1, M2	116
C — Communicating	C1, C2, C3, C4, C5, C6, C7, C8, A1, A2, G1	B2, B4, B5, B6, D1, D3, D4, D5, D6, D8, D9, D10, Q1, G2, G3, St1, P1, M1, M2, M3, M4	131
D — Applying mathematics	D1, D2, D3, D4, D5, D6, D7, D8, D9, D10, S1, Q1, F1, G2, G3, St1, P1, Fn1, M1, M2, M3	C2, S2, M4	290

Where marks are most often lost in each criterion: **B** — giving a formula with no evidence of how it was found, or generalising from too few cases. **C** — correct answers with no working, missing units, unlabelled diagrams, and conclusions that are asserted rather than explained. **D** — jumping to the calculation before defining the model, ignoring units, and never commenting on whether the answer is reasonable.

REFERENCE

Formula Bank

B

Every formula used in this question bank, grouped by topic. Learn the structure, not just the symbols.

SEQUENCES

- Arithmetic term: $u_n = a + (n - 1)d$
- Arithmetic sum: $S_n = n/2[2a + (n - 1)d]$
- Geometric term: $u_n = ar^{n1}$
- Geometric sum: $S_n = a(1 - r^n)/(1 - r)$
- Infinite sum ($|r| < 1$): $S\infty = a/(1 - r)$

QUADRATICS

- Roots: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a)$
- Vertex form: $y = a(x - h)^2 + k$
- Axis of symmetry: $x = -b/(2a)$
- Discriminant: $b^2 - 4ac$
- Completed square: $a(x + b/2a)^2 + c - b^2/(4a)$

GROWTH AND FINANCE

- Compound: A = P(1 + r)
- Compounded n times: A = P(1 + r/n)^(nt)
- Simple: A = P(1 + rt)
- Doubling time: t = ln 2 / ln(1 + r)
- Depreciation: $V = V_0(1 - r)$
- Effective rate: $(1 + r/n)^n - 1$

GEOMETRY

- Circle: $A = \pi r^2$, $C = 2\pi r$
- Arc length: $(\theta/360) \times 2\pi r$
- Sector area: $(\theta/360) \times \pi r^2$
- Cylinder: $V = \pi r^2 h$, curved area $2\pi rh$
- Cone: $V = (1/3)\pi r^2 h$, slant $l = \sqrt{(r^2 + h^2)}$
- Sphere: $V = (4/3)\pi r^3$, $A = 4\pi r^2$
- Pyramid: V = (1/3) × base area × height

TRIGONOMETRY

- sin θ = opp/hyp, cos θ = adj/hyp, tan θ = opp/adj
- Sine rule: a/sin A = b/sin B
- Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$
- Area of triangle = ½ab sin C
- $\sin^2\theta + \cos^2\theta = 1$
- sin 30° = ½, cos 30° = √3/2, tan 45° = 1

STATISTICS AND PROBABILITY

- Mean of grouped data: Σfx/Σf
- IQR = Q3 − Q1; range = max − min
- P(A or B) = P(A) + P(B) − P(A and B)
- P(A and B) = P(A) × P(B | A)
- P(B | A) = P(A and B)/P(A)
- Expected value: E(X) = Σ x P(x)

CIRCLE THEOREMS

- Angle in a semicircle = 90° (Thales)
- Angle at centre = 2 × angle at circumference
- Angles in the same segment are equal
- Cyclic quadrilateral: opposite angles sum to 180°
- Tangent radius at the point of contact
- Intersecting chords: AE × EB = CE × ED

ANGLES, BEARINGS AND UNITS

- Bearings: three figures, measured clockwise from north
- Back-bearing = bearing ± 180°
- Angles of elevation are measured from the horizontal
- 1 m^3 = 1000 litres; 1 litre = 1000 cm^3
- 1 hectare = 10 000 m^2
- Speed = distance ÷ time (km/h m/s: ÷ 3.6)

EXAM TECHNIQUE

Ten Habits That Earn Marks

C

The patterns that separate a solid answer from an excellent one in the eAssessment.

01 **Read the command word.** “Describe” needs words, “determine” needs a calculation, “justify” needs a reason, “evaluate” needs a judgement.

02 **Name the model before substituting.** A single line — “using the cosine rule” — earns the method mark even if the arithmetic goes wrong.

03 **Draw and label.** Bearing, ladder and tower problems become straightforward once the right angle and the included angle are marked.

04 **Keep exact values.** Carry π, √ and fractions to the end; round only for the final statement.

05 **State units every time.** A gradient of 6 means nothing until it is “6 dollars per hour”.

06 **Comment on the answer.** Is a population of 343 000 plausible? Is 58 cents per game fair? The interpretation is worth marks.

07 **Show the check.** Substituting back, or testing either side of a threshold, is quick and often earns a mark.

08 **Test a general rule.** After finding M(n) = 2n + 1, verify n = 4 against the diagram before moving on.

09 **For “show that”, work forwards.** Start from one side and transform it into the other — never assume the result you are asked to show.

10 **Manage the time.** Aim for roughly a mark a minute, and take the accessible parts of every question before returning to the harder ones.

Note on this answer sheet. The tasks are practice items written in the style of the MYP 5 eAssessment. Several parts are deliberately open — for example B2(e), D5(d) and D8 — so that students must *interpret* a result rather than simply calculate one. Where the printed question paper contains an inconsistency (the tree diagram in P1 and the gradient quoted in D8), the model answer says so explicitly and accepts both readings.

QUICK REFERENCE

Answer Key at a Glance

D

Every final answer in one place, for fast checking. The full method for each task is in the sections earlier in this document.

SECTION 2 · INVESTIGATING PATTERNS	
B1 Matchstick Triangles	**(a)** M = 3, 5, 7, 9, 11, 13 **(b)** The number of sticks increases by 2 each time (constant first difference). **(c)** M(n) = 2n + 1 **(d)** 41 matchsticks **(e)** No — 100 is not of the form 2n + 1 for a whole number n (99 and 101 are possible). **(f)** 3 + 2(n − 1): one shared side per new triangle gives the constant +2.
B2 Square Dot Arrays	**(a)** D: 1, 4, 9, 16, 25, 36 and P: 1, 4, 8, 12, 16, 20 **(b)** $D(n) = n^2$ and $P(n) = 4n - 4$ ($n \geq 2$) **(c)** Constant second difference of 2, so D(n) is quadratic. **(d)** $I(n) = n^2 - 4n + 4 = (n - 2)^2$ **(e)** No whole-number solution; equality would occur at n ≈ 6.83 (between Figures 6 and 7).
B3 Staircase Block Patterns	**(a)** S: 1, 3, 6, 10, 15, 21 **(b)** Quadratic sequence — constant second difference of 1. **(c)** $S(n) = \frac{1}{2}n^2 + \frac{1}{2}n$, i.e. $S(n) = n(n + 1)/2$ **(d)** 465 cubes **(e)** Disproved: T(2) = 4 ≠ 3; T(n) = S(n)(n + 2)/3, so they agree only at n = 1.
B4 Hexagonal Flower	**(a)** H: 1, 7, 19, 37, 61 (additions 1, 6, 12, 18, 24) **(b)** Ring k has 6(k − 1) dots ($k \geq 2$); ring 1 is the single centre dot. **(c)** H(n) = 1 + 3n(n − 1), derived by summing the six-sided rings. **(d)** 271 dots **(e)** 9 rings
B5 Arithmetic vs Geometric Growth	**(a)** A: 20, 35, 50, 65, 80, 95 G: 2, 4, 8, 16, 32, 64 **(b)** A(n) = 15n + 5 and $G(n) = 2^n$ **(c)** n = 14 **(d)** Straight line (constant difference) vs. upward curve (constant ratio); G overtakes A near n = 7. **(e)** n = 7; exponential growth eventually dominates any linear growth.
B6 Transforming $y = x^2$	**(a)** up 3 · left 2 · reflected in x-axis · stretched ×2 vertically; vertices (0,3), (−2,0), (0,0), (0,0) **(b)** The two expressions agree at every x (identity), shown by expanding. **(c)** $y = (x - 2)^2 + 3$, vertex (2, 3) **(d)** Left 1, vertical stretch ×2, down 4; vertex (−1, −4), minimum.
SECTION 3 · COMMUNICATING	
C1 Spot the Error: Solving an Equation	**(a)** Errors at Step 1 (incomplete expansion), Step 2 (wrong constant carried forward) and Step 4 (sign change). **(b)** x = −10 **(c)** LHS = RHS = −26
C2 Reading a Linear Graph	**(a)** $10 (the y-intercept) **(b)** Gradient = 6 dollars per hour (cost per hour of hire). **(c)** C = 6t + 10 **(d)** $40 **(e)** 6 hours
C3 Analysing a Parabola	**(a)** (0, −3); (−1, 0) and (3, 0); (1, −4) **(b)** f(x) = (x − 3)(x + 1), so x = 3 or x = −1 **(c)** $f(x) = (x - 1)^2 - 4$, vertex (1, −4) **(d)** Minimum of −4 (at x = 1), because a = 1 > 0. **(e)** x = 1, found both as the midpoint of the roots and as x = h in vertex form.
C4 Exponential Growth vs Decay	**(a)** 2 is growth (base > 1); (½) is decay (0 < base < 1). **(b)** (0, 1) for both; every $y = a^x$ has y-intercept 1 because $a^0 = 1$. **(c)** x = 4 and x = −3 **(d)** y = 0; the value becomes arbitrarily small but never zero. **(e)** No real solution, since $2^x > 0$ for all x.
C5 Understanding a Rational Function	**(a)** x = −2, because the denominator vanishes there and the numerator does not. **(b)** y = 1 (ratio of the leading coefficients). **(c)** (1, 0) and (0, −½) **(d)** x = −3.5 **(e)** f(x) > 0 for x < −2 or x > 1.
C6 The Unit Circle and Sine/Cosine	**(a)** Pythagoras on the right triangle with hypotenuse 1. **(b)** sin 30° = ½, cos 30° = √3/2, sin 60° = √3/2, cos 60° = ½ **(c)** Reflection in the y-axis keeps the sine and reverses the cosine. **(d)** Positive; sin 150° = ½

C7 Proving Thales' Theorem	**(a)** All three are radii, hence equal. **(b)** Both triangles have two radii as sides, so both are isosceles. **(c)** $\alpha + \beta = 90°$, and $ACB = \alpha + \beta$, hence 90°. **(d)** If $ACB = 90°$, then A, B, C lie on a circle with AB as diameter.
C8 Comparing Two Methods	**(a)** Terms cannot be cancelled across a subtraction — only factors of the whole expression can. **(b)** $(x - 3)/(x + 2)$, with $x \neq -3$ and $x \neq -2$ **(c)** $x = -2$ and $x = -3$
SECTION 4 · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS	
D1 Ladder Safety	**(a)** $h \approx 4.53$ m **(b)** $d \approx 2.11$ m **(c)** ≈ 4.83 m — about 0.30 m higher than at 65°. **(d)** $L \approx 4.66$ m, so a 5.0 m ladder is needed. **(e)** Vertical wall / horizontal ground / rigid ladder; allowances for footing, handhold and uneven ground.
D2 Population Growth of a City	**(a)** $P_0 = 250\,000$ and $r = 0.032$ **(b)** $\approx 343\,000$ people (342 560 to the nearest person) **(c)** 2030 ($t \approx 14.9$ years) **(d)** ≈ 22 years (2037) **(e)** Constant-rate assumption is unrealistic; a logistic model with carrying capacity is better.
D3 Is the Game Fair?	**(a)** 1/4, 1/3, 1/6, 1/4 (total 1) **(b)** $\approx$ \$3.58 per spin **(c)** $\approx$ +\$0.58 per spin — not fair; the player has the advantage. **(d)** $\approx$ \$4.58 (fair price \$3.58; school-profit price \$4.58) **(e)** $\approx$ \$200 expected
D4 Left-Handed or Right-Handed?	**(a)** 7/60 ≈ 0.117 (11.7%) **(b)** 0.10 (10%) **(c)** 4/7 ≈ 0.571 (57.1%) **(d)** Not independent (0.133 vs 0.100 vs 0.117), though only slightly dependent. **(e)** Directionally supported (13.3% vs 10.0%) but the evidence is weak — too few left-handers to conclude.
D5 Designing a Water Tank	**(a)** $h \approx 1.59$ m **(b)** $r \approx 0.89$ m **(c)** $\approx 59\,840$ CFA **(d)** $r \approx 1.0$ m (analytic optimum $r \approx 1.02$ m, $h \approx 1.53$ m), cost $\approx 58\,800$ CFA **(e)** Stability/strength, access/maintenance, footprint and local site conditions.
D6 Comparing Test Scores	**(a)** A: 42, 57, 68, 80, 95 B: 35, 60, 73, 78, 92 **(b)** B has the higher centre but the smaller IQR; A is symmetric, B is left-skewed with a low tail. **(c)** Class B — about 40% compared with about 35% in Class A. **(d)** Partly supported: B is better in the middle, but A is stronger at the top and has a higher minimum.
D7 Choosing a Savings Account	**(a)** 245 000 CFA (interest 45 000 CFA) **(b)** $\approx 243\,331$ CFA **(c)** Bank X pays more, by about 1 670 CFA. **(d)** 18 complete years **(e)** Z ≈ 241 637 CFA; choose Y over Z (about 1 690 CFA more).
D8 Study Time vs Test Score	**(a)** Strong positive linear association. **(b)** $\approx$ 24% for 6 hours and $\approx$ 39% for 11 hours **(c)** Extrapolation beyond the data range (and beyond a realistic study week). **(d)** Each extra hour of weekly study predicts roughly +3 percentage points. **(e)** e.g. prior attainment or attendance (affects both variables).
D9 Coastal Navigation	**(a)** 105° (back-bearing 220° minus bearing 115°) **(b)** AC ≈ 56 km **(c)** Bearing ≈ 084° **(d)** ≈ 3 hours 6 minutes
D10 Designing a Cone from a Sector	**(a)** $28\pi/9 \approx 9.77$ cm **(b)** $r = 14/9 \approx 1.56$ cm **(c)** $h \approx 7.85$ cm **(d)** $V \approx 19.9$ cm^3 **(e)** $\theta \approx 212°$
SECTION 5 · SEQUENCES AND SERIES IN CONTEXT	
S1 Saving Up	**(a)** 3 250 CFA in week 12 **(b)** $T(n) = 125n(n + 3)$ **(c)** 23 weeks (the total passes 70 000 CFA during that week)
S2 Bouncing Ball	**(a)** 1.6 m, 1.28 m, 1.024 m **(b)** $h_n = 2(0.8)^n$ metres **(c)** ≈ 11.45 m by the 5th impact **(d)** The distance converges to 18 m; the ball effectively stops, though the model allows infinitely many bounces.
SECTION 6 · EQUATIONS, INEQUALITIES AND FRACTIONS	
A1 Solving Equations — Mixed Practice	**(a)** $x = 6$ **(b)** $x = 22/7 \approx 3.14$, $y = 5/7 \approx 0.71$ **(c)** $x \geq -4$ **(d)** $at^2 + 2ut - 2s = 0$, giving $t = [-u \pm \sqrt{(u^2 + 2as)}]/a$
A2 Algebraic Fractions	**(a)** $(x - 5)/(x - 1)$, $x \neq 1$, $x \neq -1$ **(b)** $(5x - 1)/[(x + 1)(x - 2)]$, $x \neq -1, 2$ **(c)** $x = 1$ or $x = -3/4$
SECTION 7 · MODELLING WITH QUADRATICS	
Q1 Projectile Motion	**(a)** 2 m — the constant term in the formula. **(b)** $t \approx 3.13$ s **(c)** Maximum 13.25 m at $t = 1.5$ s **(d)** $t \approx 0.48$ s (rising) and $t \approx 2.52$ s (falling)

SECTION 8 · TRIGONOMETRIC MODELS	
F1 Tidal Heights	**(a)** 4 m **(b)** Max 6 m at t = 3 h; min 2 m at t = 9 h. **(c)** 12 hours (semidiurnal tide) **(d)** $0 \le t \le 7$ and $11 \le t \le 12$ — 8 hours in total. **(e)** Real tides are not perfectly sinusoidal or constant in amplitude.
SECTION 9 · CIRCLE THEOREMS, SOLIDS AND SCALE	
G1 Intersecting Chords	**(a)** AE · EB = CE · ED **(b)** ED = 8 **(c)** CD = 11 **(d)** CE · ED = 32; if CD AB then CD = $8\sqrt{2} \approx 11.3$ cm.
G2 Volumes of 3D Shapes	**(a)** $150\pi \approx 471.2$ cm^3 **(b)** $288\pi \approx 904.8$ cm^3 **(c)** The sphere is larger by $138\pi \approx 433.5$ cm^3. **(d)** The level drops by 8 cm. **(e)** h = 6 cm
G3 Pyramid Roof	**(a)** Slant height = 5 m **(b)** 15 m^2 per face; 60 m^2 in total **(c)** 450 000 CFA **(d)** 48 m^3
SECTION 10 · WORKING WITH GROUPED DATA	
St1 Analysing Height Data	**(a)** Roughly symmetric, single-peaked; modal class 155–160 cm. **(b)** Estimated mean = 32 005/200 = 160.0 cm **(c)** The data are nearly symmetric, so mean ≈ median. **(d)** Broadly true but overstated: 82%, not "almost all".
SECTION 11 · TREE DIAGRAMS WITHOUT REPLACEMENT	
P1 Two Marbles Without Replacement	**(a)** P(R) = 5/9, P(B) = 4/9; then 4/8 or 5/8, and 4/8 or 3/8. **(b)** 5/18 ≈ 0.278 **(c)** 5/9 ≈ 0.556 **(d)** 5/9 — the same as the first draw.
SECTION 12 · DEPRECIATION AND REDUCING BALANCE	
Fn1 Depreciating a Motorcycle	**(a)** 1 056 000 CFA after 1 year; ≈ 817 766 CFA after 3 years **(b)** V(t) = 1 200 000(0.88) **(c)** After 7 complete years **(d)** Reducing balance is higher by ≈ 153 278 CFA.
SECTION 13 · MYP 5 EASSESSMENT CHALLENGES	
M1 Planning a Concert	**(a)** 1 000 000 CFA; 400 000 CFA; 800x CFA **(b)** P(x) = 800x − 1 400 000 **(c)** 1 750 CFA per ticket **(d)** Demand falls as price rises; it is capped at the 800-seat capacity. **(e)** $\Pi(x) = -0.05x^2 + 825x - 1\,400\,000$ **(f)** x = 8 250 CFA, maximum profit ≈ 2 003 125 CFA (≈ 388 tickets).
M2 Cell Growth and Medication	**(a)** P: 200; Q: 50 **(b)** P (80 per hour vs ≈ 11.5 per hour initially). **(c)** t = 6: 680 and 200; t = 12: 1 160 and 800 **(d)** t ≈ 20 hours **(e)** About 10 hours later, at t ≈ 30 hours. **(f)** Q dominates long-term; both models ignore resource limits and drug dynamics.
M3 The Clock Tower	**(a)** Vertical tower, horizontal ground, both angles measured from the ground at C and D. **(b)** AB ≈ 15.6 m **(c)** BD ≈ 42.9 m; CD ≈ 17.9 m **(d)** ≈ 3.3° (between the lines of sight to the top and bottom of the clock) **(e)** Level ground/vertical tower; angle and distance measurement error.
M4 Rational Function Drug Model	**(a)** C(1) = 4 mg/L and C(4) = 4 mg/L **(b)** y = 0; concentration tends to zero in the long run. **(c)** t = 1.00 h and t = 4.00 h **(d)** Maximum 5 mg/L at t = 2 h **(e)** 3 hours ($1 \le t \le 4$)